A Finite Constructive Fundamental Theorem of Algebra and Rational-Function Argument Principle over the Gaussian Rationals

Signed winding certificates and rational-box approximation tracks

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Abstract

We prove a finite, computational analogue of the Fundamental Theorem of Algebra for polynomials over the Gaussian rationals. For a squarefree polynomial $f \in \mathbb{Q}(i)[Z]$ of degree $d \geq 2$, a finite algorithm returns exactly d pairwise separated refinement machines. Given a positive rational accuracy request, each machine returns a strictly refining rational box with clear boundary, algebraic winding number one, and squared diagonal below the requested bound. Canonical Gaussian-rational anchors in these boxes satisfy explicit coefficient-dependent residual estimates, while the boxes also control Hermitian projective quadrance. In degree one the zero is computed exactly. The construction does not assume that an "irrational root" is an object at all, completed or otherwise: the finite interface answering each finite request is the entire construction. The construction computes winding through signed Euclidean remainder chains and uses fraction-free Sylvester eliminants, rational Taylor bounds, bounded searches, and recursion on natural numbers.

We also derive a finite rational-function argument principle. After executable cancellation and squarefree-multiplicity decomposition, the algorithm returns a jointly separated family of zero and pole sites, exact for linear factors and given by shrinking box tracks otherwise. A site of multiplicity m has winding $+m$ for a numerator factor and $-m$ for a denominator factor; the weighted totals are the reduced degrees, and a coefficient-computed outer square has winding $\deg F_{\mathrm{red}} - \deg G_{\mathrm{red}}$.

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1 Introduction

The Fundamental Theorem of Algebra is ordinarily regarded as settled: every nonconstant polynomial over the complex numbers is said to have a complex zero. That statement, however, is formulated inside a completed complex field, and every proof route known to us uses assumptions that this paper deliberately declines. Analytic proofs begin with completed real and complex fields and use continuity there, whether through limits, the intermediate value theorem, compactness or minimization, integration, or related analytic machinery. Algebraic and topological proofs may avoid some of those devices, but still appeal to infinitary structures or existence principles such as algebraic closures, infinite extensions, global collections of roots, or choice. We do not dispute those proofs relative to their intended foundations. We ask instead what can be proved when no completed real or complex field, actual infinite object or process, analytic principle, or choice-based existence argument is admitted.

After fixing that discipline, this introductory section introduces, in dependency order, the finite definitions needed to state our constructive FTA and rational-function argument principle. The remainder of the paper then serves as a full proof of those two theorems: each later section constructs and verifies part of the finite algorithms and the certificates they return.

1.1 Statement, terminology, and constructive discipline

Under our proposed discipline the classical conclusion cannot simply be restated: to assert an arbitrary complex zero would already assume the sort of object we have declined to introduce. Our aim is the closest finite, calculation-based counterpart. From finite Gaussian-rational coefficient data we compute coefficient-checked winding certificates and finitely many independent programs. Each program answers any displayed finite accuracy request with a smaller rational box and finite proofs of refinement, clearance, winding, separation, and quantitative error bounds. No irrational root is assumed to exist behind these answers, as a completed point or as an actual infinite sequence of approximations. The finite request-and-answer interface is itself the result.

1.1.1 Related work

Choice-free constructive versions of the FTA already reach much further than a merely numerical approximation theorem. Ruitenburg [1] constructs roots over modulated Cauchy complex numbers without a choice principle. Richman [2] develops a completion suitable for a choiceless setting and identifies a monic polynomial with a completed limit multiset: a coherent method of approximately factoring the polynomial, with no advance selection of one member as a distinguished root. This is conceptually close to the present output. Indeed, each refinement procedure eventually returned here could be used as a computable name for a completed value: it answers every displayed positive rational accuracy request with finer rational data, from which one could subsequently form a Cauchy sequence and then a completed complex value. We stop before doing so: no completion is formed, equivalent names are not quotiented, and convergence to or equality of completed points is not used. Instead the finite program and each checked response are the theorem's objects. For squarefree displayed input the output also keeps the degree-many refinement procedures separated and independently refinable, rather than completing one unordered multiset. Conversely, those constructive developments admit coefficients in their completed com-

plex field, whereas our coefficient input is the narrower, fully displayed field $\mathbb{Q}(i)$. The construction in Richman's paper is presented in prose, whereas here the finite interfaces, the programs producing their certificates, and their stated guarantees are formalized together in Lean.

There is also a substantial formal lineage. Geuvers, Wiedijk, and Zwanenburg [3] formalize Kneser's constructive FTA in Coq. Their real-number structure is axiomatically a Cauchy-complete Archimedean ordered field, and their complex numbers are pairs of such reals; the formal theorem therefore retains the completed carrier and its root conclusion. Li and Paulson [4] formally verify in Isabelle/HOL that winding numbers can be evaluated through Cauchy indices, and combine that calculation with the classical argument principle to count complex roots in rectangles and half-planes. Their work gives an effective, machine-checked root-counting procedure, but its correctness theorem still uses complex-analytic winding and existing complex roots.

The closest algebraic route begins with Eisermann [5], who constructs a real-algebraic winding number from Sturm chains, derives the FTA over real-closed fields, and explains the resulting root-finding algorithm. This makes the classical theorem effective, but still assumes an ambient real-closed field and the polynomial intermediate value theorem. Perrucci and Roy [6] quantify that dependence, showing that degree-d FTA by this route needs the intermediate value theorem only through degree d^2. Perrucci and Roy [7] later correct the unrestricted auxiliary product formula, give a fully additive algebraic winding, and extend the count to rational functions, including zeros and poles on rectangle boundaries. Their result remains a theorem over $\mathbf{C} = \mathbf{R}[i]$, with $\mathbf{R}$ real closed, and counts zeros and poles already present in that algebraic closure. Our finite Cauchy-index calculations begin from the same algebraic insight but construct signed sites without adopting that carrier or its collection of points.

Finally, recursive rectangular isolation is itself classical algorithmic territory. Wilf [8] gives a global bisection algorithm that repeatedly uses argument-principle root counts to refine complex isolating rectangles. The subdivision shape is therefore not new here. What changes is its proof boundary and output: safe rational cuts, winding and clearance certificates, total independently callable refiners, and their machine-checked assembly without an analytic root-counting oracle. Across these related lines, the distinctive claim of the present work is the combined package, not the novelty of constructive approximation, algebraic winding, or box subdivision taken separately.

1.1.2 Finite geometric language

This restriction governs the paper's geometric language as well. Membership and path descriptions abbreviate calculations on displayed rational data. We use the following conventions.

- A rational interval or box is a finite record of endpoints. To say that a displayed point is inside, outside, or on its boundary means that the corresponding decidable rational equalities and inequalities hold.
- A statement such as *every* $z \in B$ means that, given any displayed z, the box-membership inequalities imply the stated conclusion. It does not construct or range over a completed collection of all points of B.
- An oriented rational path is a finite list of affine parametrizations. Its boundary image under a polynomial is the corresponding finite list of composed coordinate polynomials; it is not a completed curve or a set of all its point-values.

- A region is a rational box or cell record together with its predicates, and a family is always an explicitly finite list or finite-indexed record. Thus phrases such as *on the path*, *throughout the region*, and *the image of the boundary* assert conditional identities or inequalities for displayed rational parameters. They introduce no carrier containing all possible parameter values or points.

1.1.3 Polynomial equality

A displayed polynomial is a normalized finite coefficient list, written from highest degree to constant term. Leading zero entries are removed. We define two displayed polynomials to be equal exactly when their normalized coefficient lists agree entry by entry. For polynomials whose coefficients are themselves polynomials, and for the bivariate coordinate polynomials used below, the same definition is applied recursively to the finite coefficient data.

Consequently every polynomial identity in this paper—including Bézout, factorization, derivative, substitution, and polynomial-matrix identities—means equality of normalized coefficient lists. Evaluation by finite Horner calculation preserves such an identity, but agreement of values is never used as the definition of polynomial equality. By contrast, an expression such as $p(q) = 0$ is explicitly an equality of two displayed Gaussian-rational values after evaluation.

1.2 Machine verification

The finite machinery developed in this paper has also been implemented and verified in Lean. The accompanying formalization contains the executable procedures together with machine-checked proofs of the paper-facing guarantees; the *Formal verification* notes throughout identify the corresponding Lean declarations. This is not only a transcription of the finished prose proof, but a second presentation in which the constructions themselves can be run and their full dependency boundary inspected.

This is precisely the kind of mathematics that ought to benefit from machine verification. Every object in the construction is finite data; every search has a displayed finite bound; every recursive call decreases an explicit natural measure; and every certificate is checked by finite algebra, integer arithmetic, or rational comparison. There is no appeal to a picture, a limiting object, or an unnamed witness whose properties must be taken on trust. A proof assistant can therefore check the argument at the same level at which the theorem is stated. Lean also permits the programs and their correctness proofs to share one environment, so the formalization checks not only that the certificates imply the advertised conclusions, but that the finite procedures producing them are total and executable. Machine verification is consequently part of the constructive discipline of this paper, not decorative evidence added after the mathematics was finished.

The consequences of that choice for Lean, Mathlib, and the proof's audited axiom boundary are discussed in appendix A.

1.3 Definitions and theorem statements

The motivational and foundational preliminaries are now complete. We begin the chain of finite definitions needed to state the two main theorems. The definitions are arranged in dependency order and culminate first in the finite constructive FTA, and then in the finite

rational-function argument principle. This subsection deliberately prioritizes reaching those statements quickly: it retains the data needed to read them, while deferring most intuition, motivation, and derivation to the later section where each construction is first used. The short links following each group point to that deferred development.

Polynomial names are role-based. The letter f denotes the polynomial currently under the main winding, isolation, or FTA argument, whereas p denotes an arbitrary polynomial supplied to a reusable bound, auxiliary algorithm, or preprocessing step. Letters such as g, h denote temporary comparison polynomials, GCDs, or factors, and U, V denote displayed Bézout cofactors. In the rational-function statements, F, G are the original numerator and denominator, F_0, G_0 their reduced cofactors, P, Q a generic polynomial pair, and indexed p_j, q_k individual squarefree factors.

1.3.1 Polynomial input certificates

Definition 1.1 (squarefree certificate)**.** *For $f \in \mathbb{Q}(i)[Z]$, a* squarefree certificate *is a checked nonzero constant output of the finite Euclidean GCD computation on f, f'. Rescaling the computed Bézout cofactors gives displayed $U, V \in \mathbb{Q}(i)[Z]$ satisfying*

$$U f + V f' = 1.$$

Formal verification:

`Purelib.Polynomial.Squarefree.SquarefreeCertificate;Purelib.Polynomial.Squarefree.CoprimeCertificate.identity`.

Its quantitative role is developed in section 2.

1.3.2 Finite metric and box data

Definition 1.2 (Gaussian-rational quadrance)**.** *For a displayed Gaussian rational $z = x + iy$, its* quadrance *is*

$$\mathrm{N}(z) = x^2 + y^2 \in \mathbb{Q}_{\geq 0}.$$

Formal verification:

`Purelib.Data.Gaussian.normSq`.

Definition 1.3 (Hermitian projective quadrance)**.** *For displayed $z, w \in \mathbb{Q}(i)$, their* Hermitian projective quadrance *is*

$$\chi^2(z, w) = \frac{\mathrm{N}(z - w)}{(1 + \mathrm{N}(z))(1 + \mathrm{N}(w))}. \tag{1}$$

This is the Gaussian-Hermitian affine analogue of Wildberger's projective quadrance [9]; here it is used only as a rational error measure.

Formal verification:

`Purelib.Polynomial.Geometry.Chordal.affineChordalSq`.

Definition 1.4 (rational box)**.** *A* rational box *is a tuple*

$$B = (x_-, x_+, y_-, y_+) \in \mathbb{Q}^4, \qquad x_- < x_+, \quad y_- < y_+.$$

A point $z = x + iy$ belongs to B, written $z \in B$, when

$$x_- \le x \le x_+, \qquad y_- \le y \le y_+.$$

The squared diagonal *of B is*

$$\operatorname{diag}^2(B) = (x_+ - x_-)^2 + (y_+ - y_-)^2.$$

One box is contained in another, written $D \subseteq B$, when both of its coordinate intervals are contained; this symbol allows equality. If $D = (u_-, u_+, v_-, v_+)$, write $D \subset B$ and call D a strict box refinement *of B when*

$$x_- < u_-, \qquad u_+ < x_+, \qquad y_- < v_-, \qquad v_+ < y_+.$$

Thus $\subset$ always denotes strict coordinatewise containment between rational boxes in this paper, whereas $\subseteq$ denotes containment that may include equality.

Formal verification:

`Purelib.Polynomial.Geometry.Boxes.Box;Purelib.Polynomial.Geometry.Boxes.Box.Proper;Purelib.Polynomial.Geometry.Boxes.Box.Contains;Purelib.Polynomial.Geometry.Boxes.Box.Refines;Purelib.Polynomial.Geometry.Boxes.Box.StrictInterior;Purelib.Polynomial.Geometry.Boxes.Box.diagonalSq`.

Definition 1.5 (box anchor and opposite corner)**.** *For the rational box B of definition 1.4, put*

$$q_B = x_- + iy_-, \qquad v_B = x_+ + iy_+.$$

Call q_B the canonical rational anchor *of B.*

Formal verification:

`Purelib.Polynomial.Geometry.Boxes.Box.anchor;Purelib.Polynomial.Geometry.Chordal.oppositeCorner`.

Definition 1.6 (coefficient isolation square)**.** *Let*

$$f(Z) = a_d Z^d + \sum_{k=0}^{d-1} a_k Z^k, \qquad d \ge 1, \quad a_d \ne 0.$$

Compute the two positive rational quantities

$$C_f = d \sum_{k=0}^{d-1} 2^k \,\mathrm{N}(a_k)\,, \qquad R_f = \max\left\{1, \frac{C_f}{\mathrm{N}(a_d)} + 1\right\}. \tag{2}$$

The coefficient isolation square *of f is the rational box*

$$K_f = (-R_f, R_f, -R_f, R_f).$$

Formal verification:

`Purelib.RootIsolation.Winding.Global.denseLargeSquareRadius;Purelib.RootIsolation.Winding.Global.largeSquareBox`.

Definition 1.7 (ambient box)**.** *An* ambient box *is a fixed rational box A together with the stored containment certificates $B \subseteq A$ for the working boxes, supplying one common domain for coefficient bounds. In the main polynomial theorem $A = K_f$.*

Definition 1.8 (common coefficient isolation square)**.** *For two nonzero polynomials P, Q, let $R_{P,Q}$ be the larger of the radii given by (2) for their nonconstant members. A constant imposes no radius condition; if both polynomials are constant, put $R_{P,Q} = 1$. Their* common coefficient isolation square *is*

$$K_{P,Q} = (-R_{P,Q}, R_{P,Q}, -R_{P,Q}, R_{P,Q}).$$

For box records, A denotes a fixed ambient box, B a generic current or parent box, C an intermediate child selected during subdivision, and D a returned refined box. The symbols $K_f, K_R, K_{P,Q}$ are reserved for coefficient-computed outer isolation squares rather than arbitrary boxes. Displayed Gaussian-rational samples are denoted by z, w, q; q_B and v_B are respectively the anchor and opposite corner of B.

Definition 1.9 (quadrance-Lipschitz certificate)**.** *Let $p \in \mathbb{Q}(i)[Z]$ and let A be a rational box. A* quadrance-Lipschitz certificate *on A is a displayed rational $L > 0$ together with a finite proof that, for any displayed $z, w \in \mathbb{Q}(i)$, the two box-membership predicates imply*

$$\mathrm{N}(p(z) - p(w)) \le L\,\mathrm{N}(z - w)\,. \tag{3}$$

We write $L_1(p, A)$ for the particular coefficient-computed certificate constructed in section 2 from the finite divided-difference identity (7).

Formal verification:

`Purelib.Polynomial.Bounds.denseLipschitzBound;Purelib.Polynomial.Bounds.normSq_eval_sub_eval_le_denseLipschitzBound`.

The metric estimates are proved in section 2, and the degree winding of the coefficient isolation square in section 6.

1.3.3 Certified states and finite tracks

Definition 1.10 (boundary-clearance certificate)**.** *For $f \in \mathbb{Q}(i)[Z]$ and a rational box B, a* boundary-clearance certificate *is a displayed rational $m_B > 0$ together with four finite proofs that*

$$m_B \le \mathrm{N}(f(z))$$

on the four rational edge parametrizations of B.

Formal verification:

`Purelib.Polynomial.Geometry.BoxClearance.BoundaryClearance`.

Definition 1.11 (box winding)**.** *Let $P, Q \in \mathbb{Q}(i)[Z]$, and let a rational box B carry boundary-clearance certificates for both polynomials. On each oriented affine edge, represent the direction of the displayed quotient $P(z)/Q(z)$ by the Gaussian-rational coordinate pair*

$$P(z)\overline{Q(z)}.$$

The four finite signed Cauchy–Sturm edge calculations, realized explicitly in section 4, give the integer $\mathrm{Wind}_4(P/Q; B)$. *Its conventional normalization is*

$$\mathrm{Wind}_{P/Q}(B) = \mathrm{Wind}_4(P/Q; B)/4.$$

For a polynomial f, *the case* $P = f, Q = 1$ *is denoted* $\mathrm{Wind}_4(f; B)$ *and* $\mathrm{Wind}_f(B)$. *The finite constructions below use* Wind_4*; the normalized notation is reserved for the headline statements. No winding is assigned without the required boundary-clearance certificates.*

Formal verification:

`Purelib.RationalFunctions.Winding.rationalQuadrupledWinding;Purelib.RationalFunctions.Winding.rationalBoxWinding;Purelib.RootIsolation.Winding.quadrupledBoxWinding;Purelib.RootIsolation.Winding.boxWinding`.

Definition 1.12 (winding-one certificate)**.** *A* winding-one certificate *for* f *and* B *consists of its four finite edge-chain calculations, their endpoint signs, and the integer check*

$$\mathrm{Wind}_4(f; B) = 4.$$

A finite root state also carries the clearance certificate of definition 1.10.

Formal verification:

`Purelib.RootIsolation.Isolation.ShrinkingTrack.WindingOneState;Purelib.RootIsolation.Winding.quadrupledBoxWinding`.

Definition 1.13 (finite root state)**.** *Let* $f \in \mathbb{Q}(i)[Z]$. *A* finite root state *consists of*

$$(B, \mathcal{C}, \mathcal{W}),$$

where B *is a rational box,* $\mathcal{C}$ *is a boundary-clearance certificate, and* $\mathcal{W}$ *is a winding-one certificate. Its displayed sample is the anchor* q_B *of definition 1.5; the defining rational inequalities give* $q_B \in B$.

Formal verification:

`Purelib.RootIsolation.Isolation.ShrinkingTrack.WindingOneState`.

Definition 1.14 (refinement of root states)**.** *A finite root state* σ' refines *a state* σ *when its box* B' *satisfies* $B' \subseteq B$ *in the sense of Definition 1.4. The state refinement is* strict *when* $B' \subset B$. *In either case its refinement certificate is the finite list of coordinate inequalities witnessing the indicated box relation.*

From this point σ, σ' denote complete certified state records; B, B' denote their box components when no other state data is needed.

Definition 1.15 (root-approximation track, RAT)**.** *A* root-approximation track*, abbreviated* RAT*, for* f *consists of a finite initial root state* σ_0 *and a deterministic refinement rule. Given the current root state* σ *and a displayed* $\varepsilon \in \mathbb{Q}_{>0}$*, the rule terminates and returns one new root state*

$$(\sigma, \varepsilon) \longmapsto \sigma' = \mathrm{refine}(\sigma, \varepsilon).$$

The returned state σ' *is a strict refinement of* σ*, and its box* B' *satisfies*

$$\mathrm{diag}^2(B') \le \varepsilon^2.$$

Formal verification:

`Purelib.RootIsolation.Isolation.ShrinkingTrack.WindingOneTrack;Purelib.RootIsolation.Isolation.ShrinkingTrack.ofInitial`.

Definition 1.16 (finite observations and separation)**.** *An* exact terminal *for a polynomial p is a displayed $q \in \mathbb{Q}(i)$ with exact evaluation $p(q) = 0$. A* finite observation *is either an exact terminal or a RAT. At any finite stage it has a rational* support record

$$S = (x_-^S, x_+^S, y_-^S, y_+^S).$$

For an exact terminal $q = a + ib$, put $S = (a, a, b, b)$. For a RAT, make a finite call and use its returned box as S.

Two finite observations are separated *when we can produce support records S, T and a displayed $g \in \mathbb{Q}_{>0}$ satisfying one of*

$$x_+^S + g \le x_-^T, \quad x_+^T + g \le x_-^S,$$
$$y_+^S + g \le y_-^T, \quad y_+^T + g \le y_-^S.$$

A finite family is jointly separated *when such a certificate is stored for every pair of distinct observations. Refining a RAT preserves every stored witness.*

Formal verification:

`Purelib.Polynomial.Geometry.Boxes.Box.Separated;Purelib.Polynomial.Geometry.Boxes.Box.SeparationCertificate;Purelib.Polynomial.Geometry.Boxes.Box.separationCertificate;Purelib.Polynomial.Geometry.Boxes.Box.Separated.of_refines;Purelib.RationalFunctions.Sites.PositiveSideInput.ObservationsSeparated;Purelib.RationalFunctions.Sites.PositiveSideInput.separateObservations;Purelib.RationalFunctions.SignedMultiplicity.TrackPairSeparation`.

Certified states, separation, and total tracks are constructed in sections 7 to 9.

1.3.4 The finite theorem

Theorem 1.17 (finite constructive FTA)**.** *Let $f \in \mathbb{Q}(i)[Z]$ have degree $d \ge 1$, nonzero leading coefficient, and a squarefree certificate. A finite algorithm returns one of the following.*

(i) If $d = 1$, it returns a directly computed exact terminal $q \in \mathbb{Q}(i)$ for f.
(ii) If $d \ge 2$, it returns a list of exactly d RATs.

In the second case the returned family has four certified properties.

(a) Separation. *Its initial boxes are pairwise separated in the sense of Definition 1.16, viewing each RAT as a finite observation, with displayed positive rational gaps.*
(b) Refinement. *Every call terminates and returns the strict refinement required by Definition 1.15, including the anchor q_B of Definition 1.5.*
(c) Quantitative control. *Let K_f be the coefficient isolation square of Definition 1.6. Every returned box satisfies $B \subseteq K_f$. The quadrance-Lipschitz certificate of Definition 1.9 gives*

$$\mathrm{N}(f(q_B)) \le L_1(f, K_f)\,\mathrm{diag}^2(B).$$

For every displayed $z \in B$, one also has

$$\begin{aligned}\chi^2(q_B, z) &\le \operatorname{diag}^2(B),\\ \mathrm{N}(f(z)) &\le 4L_1(f, K_f)\operatorname{diag}^2(B),\\ \mathrm{N}(f(q_B)) &\le L_1(f, K_f)(1+\mathrm{N}(q_B))(1+\mathrm{N}(v_B))\chi^2(q_B, v_B).\end{aligned}$$

In particular a request with $\operatorname{diag}^2(B) \le \varepsilon^2$ bounds both quantities by $\chi^2(q_B, z) \le \varepsilon^2$ and $\mathrm{N}(f(q_B)) \le L_1(f, K_f)\varepsilon^2$. Consecutive anchors $q_B, q_{B'}$ satisfy $\mathrm{N}(q_{B'} - q_B) \le \operatorname{diag}^2(B)$ for the earlier box.

(d) Finite maximality. *No finite family of pairwise separated RATs supported in the same coefficient isolation square K_f has length greater than d.*

The algorithm uses only finite arithmetic, bounded searches over displayed finite lists, and recursion on natural numbers.

Formal verification:

`Purelib.RootIsolation.Isolation.BoxFTA.provePositive;Purelib.Papers.RFAP.Paper.isolateByBoxes`.

The proof is completed in section 9.

1.3.5 Rational-function data and statement

The rational-function theorem uses the same polynomial tracks. The extra records below say exactly what cancellation, multiplicity, sign, and cross-side separation mean.

Definition 1.18 (reduced quotient presentation)**.** *Let $F, G \in \mathbb{Q}(i)[Z]$ be displayed nonzero polynomials. A* reduced quotient presentation *consists of five displayed polynomials*

$$H, F_0, G_0, U, V \in \mathbb{Q}(i)[Z]$$

and the identities

$$F = HF_0, \qquad G = HG_0,$$
$$UF_0 + VG_0 = 1,$$

together with the finite-division degree checks

$$\deg H + \deg F_0 = \deg F, \qquad \deg H + \deg G_0 = \deg G.$$

Formal verification:

`Purelib.RationalFunctions.Reduction.ReducedPair;Purelib.RationalFunctions.Reduction.reduce`.

Definition 1.19 (finite squarefree-multiplicity presentation)**.** *For a displayed nonzero polynomial P, a* finite squarefree-multiplicity presentation *is an identity*

$$P = u\prod_{j=1}^{r} p_j^{m_j},$$

where $u \in \mathbb{Q}(i)^\times$, every p_j has positive degree and a squarefree certificate, every m_j is a displayed positive integer, and every pair of distinct factors p_j, p_k carries displayed $A_{jk}, B_{jk} \in \mathbb{Q}(i)[Z]$ with the Bézout identity

$$A_{jk}p_j + B_{jk}p_k = 1.$$

The record also contains the checked degree identity

$$\sum_{j=1}^{r} m_j \deg p_j = \deg P.$$

A nonzero constant has $r = 0$ and the empty product.

Formal verification:

`Purelib.Polynomial.Factorization.Decomposition.positiveMultiplicityEntries;Purelib.Polynomial.Factorization.Decomposition.positiveMultiplicityDegree;Purelib.Polynomial.Multiplicity.PositiveEntry`.

Definition 1.20 (signed rational-function site)**.** *Fix one factor p_j in a presentation of a reduced numerator or denominator. A* signed site *records its positive multiplicity m_j, the sign $+$ for a numerator factor or $-$ for a denominator factor, and a finite observation in the sense of Definition 1.16, with the following additional data.*

(i) If $\deg p_j = 1$, the observation is an exact terminal q, together with a certificate that the opposite reduced cofactor does not vanish at q.

(ii) If $\deg p_j \geq 2$, the observation is one of the $\deg p_j$ RATs for p_j, together with finite boundary certificates showing that the full reduced quotient has winding $+4m_j$ on its returned boxes in the numerator case, or $-4m_j$ in the denominator case.

Formal verification:

`Purelib.RationalFunctions.Sites.PositiveSideInput.FactorObservation;Purelib.RationalFunctions.SignedMultiplicity.TrackSiteInput`.

The rational-function theorem uses the box winding of Definition 1.11.

Theorem 1.21 (finite constructive rational-function action principle)**.** *For displayed $F, G \in \mathbb{Q}(i)[Z]$ with $G \neq 0$, a total finite algorithm returns one of the following.*

(a) If $F = 0$, it returns the exact zero-numerator certificate. It assigns no boundary winding to the identically zero quotient.

(b) If $F \neq 0$, it returns a reduced quotient presentation in the sense of Definition 1.18; finite squarefree-multiplicity presentations of F_0 and G_0 in the sense of Definition 1.19; and all their signed sites in the sense of Definition 1.20. The sites are jointly separated as finite observations in the sense of Definition 1.16.
The degree-weighted positive numerator multiplicities total $\deg F_0$, and the degree-weighted absolute denominator multiplicities total $\deg G_0$, as in Definition 1.19. The common coefficient isolation square K_{F_0,G_0} of Definition 1.8 is boundary-clear for both cofactors and satisfies

$$\mathrm{Wind}_{F_0/G_0}(K_{F_0,G_0}) = \deg F_0 - \deg G_0$$

in the sense of Definition 1.11.

For every higher squarefree factor, any competing winding-one RAT supported in that factor's coefficient isolation square has a deterministically computed owner among the returned tracks. Consequently every pairwise-separated competing family for that factor has cardinality at most its degree.

Formal verification:

`Purelib.RationalFunctions.Sites.prove;Purelib.Papers.RFAP.Paper.rationalFunctionArgumentPrinciple`.

The multiplicity presentations are constructed in section 10; the remaining signed-site construction and the proof of Theorem 1.21 are completed in section 11.

1.4 Proof-route glossary

These entries form a reader's map, not a second set of formal definitions: their dependency order makes the direction of the proof visible.

Displayed data. An object is *displayed* when it is given by finite rational or integer data on which the stated calculation can actually be run. A displayed polynomial is its normalized finite coefficient list; a displayed point has Gaussian-rational coordinates.

Certificate. A *certificate* is displayed data together with finite checks implying the claimed property. Later algorithms consume the certificate itself and return further certificates.

Polynomial identity. A *polynomial identity* is a certificate that the normalized coefficient lists agree, as fixed in section 1.1.3. Substitution may turn such an identity into an identity of displayed values, but value agreement is not polynomial equality.

Squarefree certificate. The Bézout identity $Uf + Vf' = 1$ is the *squarefree certificate*. Finite coefficient bounds for U and V turn it into quantitative squarefreeness: if the displayed value of f is small in a box, the displayed value of f' there cannot also be small. The resulting conditional positive lower bound for $\mathrm{N}(f')$ is the *derivative margin* used by the Newton step.

Quadrance, projective quadrance, and coefficient bound. For a displayed Gaussian rational $z = x + iy$, we write $\mathrm{N}(z) = x^2 + y^2 \in \mathbb{Q}_{\geq 0}$ for its norm, which will also refer to as its *quadrance*. A *Hermitian projective quadrance* is the normalized rational error

$$\chi^2(z, w) = \frac{\mathrm{N}(z - w)}{(1 + \mathrm{N}(z))(1 + \mathrm{N}(w))};$$

it is the bounded metric quantity controlled by the rational boxes. A *coefficient bound* is a rational number computed by a finite fold over a polynomial's coefficients and proved to bound a stated quadrance on a rational interval or box. A *quadrance-Lipschitz certificate* packages such bounds for $f(z) - f(w)$ and its first-order Taylor remainder. A *Horner majorant* is a coefficient bound computed by running a rational Horner recursion on the displayed coefficient list; it is not a maximum over an interval or box.

Rational box, anchor, and ambient box. A *rational box* is four rational endpoints; its *anchor* is its lower-left Gaussian-rational corner, and its *squared diagonal* is the sum of the squares of its two side lengths. An *ambient box* is a fixed outer box in which coefficient bounds have been certified and inside which working boxes are refined. A *support record*

is the same four-endpoint shape used uniformly for separation: a RAT supplies one of its boxes, while an exact terminal supplies the degenerate box at its displayed value.

Boundary clearance. A box is *boundary-clear* for a polynomial when a certificate gives one positive rational lower bound for its quadrance on all four displayed edge parametrizations. Clearance is what permits the winding calculation below and is preserved when a cleared edge is reused by subdivision. Combining clearance with coefficient bounds gives a *protected boundary radius*: within that rational coordinate distance, a new boundary still cannot make f vanish. A *protected parent* is a proper boundary-clear box together with its refinement into a fixed ambient box; it retains exactly the finite local data needed for residual-driven Newton extraction, whether or not its winding is recorded.

Signed remainder chain and Cauchy index. A *signed remainder chain* is the finite Euclidean remainder program for two rational coordinate polynomials. Its endpoint sign variations give their *Cauchy index*, a signed count of crossing events on a displayed interval.

Box winding. The *box winding* Wind_4 is the integer obtained by adding the four Cauchy indices on a boundary-clear box. The subscript records the retained factor of four; a *winding-one* box has $\mathrm{Wind}_4 = 4$.

Eliminant, safe cut, and subdivision. An *eliminant* is a displayed nonzero rational polynomial whose nonzero value at a coordinate produces a boundary-clearance certificate for the corresponding vertical or horizontal line. A bounded scan finds such a *safe cut*. Safe cuts divide a boundary-clear box into a finite grid of boundary-clear boxes, and winding additivity distributes the parent's box winding among those children.

Leading–tail dominance, global winding budget, and local extraction. Outside the *coefficient isolation square*, the polynomial's displayed *leading monomial* dominates its *lower tail*; their explicit quadrance gap is the *leading–tail dominance margin*. On the boundary of that square this gives the *global winding budget* $\mathrm{Wind}_4 = 4 \deg f$. *Local extraction* repeatedly subdivides a child with nonzero winding and uses finite Newton and Taylor certificates to obtain a *winding-one root state*: a rational box carrying both boundary-clearance and winding-one certificates.

Newton centre, Newton band, and finite Rouché certificate. When the derivative margin proves $f'(a) \neq 0$, the *Newton centre* is the displayed Gaussian rational $c = a - f(a)/f'(a)$. At radius $r > 0$, a *Newton band* is a rational rectangle whose left, right, bottom, and top sides are chosen respectively at coordinate distances between r and $2r$ from c. Safe-cut searches choose its four sides. A Taylor *dominance certificate* proves on the complete boundary that the affine Newton term is larger than the nonlinear remainder; the finite Rouché rule then transfers winding four from the affine term to the selected band.

Root-approximation track (RAT) and finite observation. A *RAT* packages a winding-one root state with a total refinement program. A request for a displayed accuracy returns a strictly smaller boundary-clear, winding-one box and its anchor. A *finite observation* is either such a RAT or an *exact terminal*, which is a displayed Gaussian-rational point whose polynomial value is exactly zero.

Residual. The polynomial *residual* at a displayed q is the rational quadrance $\mathrm{N}(f(q))$. Small winding-one boxes give small residuals; with a protected parent, a sufficiently small residual conversely constructs a nearby RAT.

Separation, ownership, and finite maximality. *Separation* is a positive rational coordinate gap between the support boxes of two observations. Given output boxes $B_1, \dots, B_d$ and a new refined box D, an *ownership certificate* gives the unique index j, with $1 \le j \le d$, for which $D \subset B_j$; the jth returned RAT is called the *owner*. For a separated family these assignments are distinct, giving *finite maximality*: there can be no more competing RATs than output boxes.

Squarefree preprocessing, multiplicity presentation, and signed site. *Squarefree preprocessing* is the terminating finite gcd recursion that splits a displayed polynomial into squarefree occurrences and then folds overlaps into pairwise-coprime factors. Its output is a *multiplicity presentation*: a polynomial identity built from those factors and positive integer powers. A *signed site* is an exact terminal or RAT for one such factor, labelled by its multiplicity and by $+$ for a numerator or $-$ for a denominator.

Reduced quotient and rational-function action principle. A *reduced quotient* cancels the displayed common factor of numerator and denominator and certifies the remaining pair as coprime. Applying the polynomial construction to its multiplicity presentations produces jointly separated signed sites; adding their signed winding contributions is the finite rational-function action principle.

Protected quotient residual. At a numerator site for a reduced quotient P/Q, a finite Bézout certificate, a source RAT, and its current state form a *local quotient context*. Its finite refinement gives a *protected state* on which the opposite polynomial Q has displayed positive lower and finite upper quadrance bounds. The *quotient residual* is $\mathrm{N}(P(q)/Q(q))$. At a denominator site the roles are exchanged: the shrinking quantity is the reciprocal residual $\mathrm{N}(Q(q)/P(q))$, and the protected opposite polynomial is P.

2 Finite polynomial bounds over $\mathbb{Q}(i)$

This section computes coefficient bounds for displayed polynomials on rational boxes. These bounds turn squarefree certificates into the boundary-clearance and local-extraction certificates used later.

The estimates in this section have three simple jobs. The coefficient L_1 controls how much a polynomial value can change across a small box; L_2 controls the error made by replacing the polynomial there with its affine Taylor expression; and the pair (τ, η) turns squarefreeness into the implication "small value, non-small derivative." Later constructions repeatedly choose a rational scale at which these three controls are simultaneously valid. The formulas are lengthy because every constant is computed from coefficients, not because they introduce three different notions of approximation.

Lemma 2.1 (finite quadrance algebra)**.** *For displayed $z, w, u, v \in \mathbb{Q}(i)$,*

$$\mathrm{N}(zw) = \mathrm{N}(z)\,\mathrm{N}(w)\,, \qquad \mathrm{N}(u+v) \le 2\big(\mathrm{N}(u) + \mathrm{N}(v)\big).$$

For every displayed finite list $u_1, \dots, u_m \in \mathbb{Q}(i)$,

$$\mathrm{N}(u_1 + \cdots + u_m) \le m \sum_{j=1}^{m} \mathrm{N}(u_j)\,. \tag{4}$$

Proof. Expand the two rational coordinates for the product identity. The two-term inequality follows after subtracting the left side from the right: the difference is a sum of two rational squares. For the finite-list bound, apply

$$(r_1 + \cdots + r_m)^2 \le m \sum_{j=1}^{m} r_j^2$$

separately to the real and imaginary coordinate lists and add the two inequalities. The difference in the displayed scalar inequality is a finite sum of squares. □

Formal verification:

`Purelib.Data.Gaussian.normSq_mul;Purelib.Data.Gaussian.normSq_add_le_two; Purelib.LinearAlgebra.Norm.normSq_gaussianSum_le`.

Lemma 2.2 (rational-box norm bound)**.** *For a rational box B, put*

$$R_B = 2\max\{x_-^2, x_+^2, y_-^2, y_+^2\}.$$

Every displayed $z \in B$ satisfies $\mathrm{N}(z) \le R_B$.

Proof. The rational interval inequalities for the two coordinates give

$$\mathrm{Re}(z)^2 \le \max\{x_-^2, x_+^2\}, \qquad \mathrm{Im}(z)^2 \le \max\{y_-^2, y_+^2\}.$$

Both right-hand sides are at most the maximum of all four endpoint squares. Adding the two bounds proves the claim. □

Formal verification:

`Purelib.Polynomial.Geometry.Boxes.Box.normSqBound;Purelib.Polynomial. Geometry.Boxes.Box.normSq_le_normSqBound`.

Quadrance is squared Euclidean magnitude kept rational, so no square root is required. The Hermitian projective quadrance used here is the Gaussian-Hermitian analogue, on the affine chart, of Wildberger's algebraic projective separation of homogeneous directions: z, w represent $[z : 1]$ and $[w : 1]$. Its displayed rational value is the error measure used below.

Proposition 2.3 (box calibration of Hermitian projective quadrance)**.** *For a rational box B and every displayed $z, w \in B$,*

$$\chi^2(z, w) \le \mathrm{N}(z - w) \le \mathrm{diag}^2(B). \tag{5}$$

At the distinguished opposite corners the exact comparison is

$$\mathrm{diag}^2(B) = (1 + \mathrm{N}(q_B))(1 + \mathrm{N}(v_B))\chi^2(q_B, v_B). \tag{6}$$

Proof. The denominator in equation (1) is at least one. Membership in B bounds the real and imaginary coordinate differences by the two side lengths, proving equation (5). For q_B, v_B, the numerator $\mathrm{N}(q_B - v_B)$ is exactly $\mathrm{diag}^2(B)$; clearing the explicitly positive denominator gives equation (6). □

Formal verification:

`Purelib.Polynomial.Geometry.Chordal.affineChordalSq_le_box_diagonal; Purelib.Polynomial.Geometry.Chordal.diagonalSq_eq_cornerWeight_mul_affineChordalSq`.

Lemma 2.4 (finite Horner evaluation bound)**.** *Let $p \in \mathbb{Q}(i)[Z]$ and let B be a rational box. Write the displayed high-to-low coefficient list of p as $[a_n, \ldots, a_0]$. Starting from $M_0 = 0$, perform the finite recurrence*

$$M_{k+1} = 2\big(M_k R_B + \mathrm{N}(a_{n-k})\big).$$

Then every z satisfying the predicate of B obeys $\mathrm{N}(p(z)) \le M_{n+1}$. Simultaneously folding $S_{k+1} = 2(S_k R_B + M_k)$ gives the finite divided-difference coefficient used below. Both outputs are rational computations on the finite coefficient list.

Proof. Induct over the Horner fold and use $\mathrm{N}(uv) = \mathrm{N}(u)\,\mathrm{N}(v)$ and $\mathrm{N}(u+v) \le 2(\mathrm{N}(u) + \mathrm{N}(v))$ at each step. Applying the same induction to two Horner evaluations gives the displayed S_k recurrence. □

Formal verification:

`Purelib.Polynomial.Bounds.denseHornerNormBounds;Purelib.Polynomial.Bounds.normSq_eval_le_denseEvalNormBound`.

The same finite fold has the following one-variable rational specialization. We name it separately because it is the form used below for Bézout cofactors on parameter intervals.

Definition 2.5 (rational-interval Horner majorant)**.** *Let $a \le b$ be displayed rationals, write $I = [a, b]$, and put*

$$R_I = \max\{a^2, b^2\}.$$

For $W \in \mathbb{Q}[T]$ with displayed high-to-low coefficient list $[w_n, \ldots, w_0]$, start from $H_0 = 0$ and perform the finite recurrence

$$H_{k+1} = 2(H_k R_I + w_{n-k}^2).$$

The resulting rational

$$\mathcal{H}_I(W) := H_{n+1}$$

is the Horner majorant *of W on I. It is computed from the finite coefficient list and the two endpoints; it is not a maximum taken over the parameters satisfying the interval predicate.*

Lemma 2.6 (rational-interval Horner bound)**.** *For the data of definition 2.5 and every displayed $t \in \mathbb{Q}$ satisfying $a \le t \le b$,*

$$W(t)^2 \le \mathcal{H}_I(W).$$

The majorant is a nonnegative rational computed by one finite fold over the displayed coefficient list.

Proof. The endpoint inequalities give $t^2 \le R_I$. Let $E_0 = 0$ and $E_{k+1} = E_k t + w_{n-k}$ be the corresponding Horner evaluation. If $E_k^2 \le H_k$, then

$$\begin{aligned} E_{k+1}^2 &\le 2(E_k^2 t^2 + w_{n-k}^2) \\ &\le 2(H_k R_I + w_{n-k}^2) = H_{k+1}. \end{aligned}$$

Finite induction gives $W(t)^2 = E_{n+1}^2 \le H_{n+1}$. The same induction, starting from $H_0 = 0$, proves $H_k \ge 0$ at every step. □

Formal verification:

`Purelib.Polynomial.Geometry.PathClearance.evalSqBoundStep;Purelib.Polynomial.Geometry.PathClearance.evalSqBound;Purelib.Polynomial.Geometry.PathClearance.eval_sq_le_evalSqBound`.

Lemma 2.7 (explicit geometric depth)**.** *For displayed rationals $w \ge 0$ and $\varepsilon > 0$, define*

$$\mathrm{geometricDepth}(w, \varepsilon) := \left\lceil \frac{2w}{\varepsilon} \right\rceil \in \mathbb{N}.$$

Then

$$\left(\frac{2}{3}\right)^{\mathrm{geometricDepth}(w,\varepsilon)} w \le \varepsilon.$$

Finite minima of displayed positive rationals remain positive.

Proof. Put $N = \mathrm{geometricDepth}(w, \varepsilon)$. The constructive rational ceiling gives $2w/\varepsilon \le N$. The finite Bernoulli induction $(3/2)^N \ge 1 + N/2$ then implies $(2/3)^N w \le \varepsilon$. Positivity of a finite minimum is proved by induction on the displayed list. □

Formal verification:

`Purelib.RootIsolation.Isolation.SafeCandidates.geometricDepth;Purelib.RootIsolation.Isolation.SafeCandidates.geometricDepth_sufficient`.

Lemma 2.8 (finite divided-difference bound)**.** *Let $p \in \mathbb{Q}(i)[Z]$ and let B be a rational box. Write $p(Z) = \sum_{j=0}^{n} a_j Z^j$. The finite identity*

$$p(z) - p(w) = (z - w) D_p(z, w), \qquad D_p(z, w) = \sum_{j=1}^{n} a_j \sum_{r=0}^{j-1} z^{j-1-r} w^r \tag{7}$$

computes a rational $L_1(p, B) > 0$ such that, whenever $z, w \in B$,

$$\mathrm{N}(p(z) - p(w)) \le L_1(p, B)\, \mathrm{N}(z - w) \tag{8}$$

and hence supplies the certificate of Definition 1.9.

Proof. Factor $z^j - w^j$ as $(z-w)\sum_{r=0}^{j-1} z^{j-1-r} w^r$ and sum the finitely many monomial identities to obtain (7). The resulting D_p is a displayed finite bivariate polynomial. Apply the Horner bounds of lemma 2.4 in the two variables using the box norm bound of lemma 2.2, and combine the result with norm multiplicativity from lemma 2.1. □

Formal verification:

`Purelib.Polynomial.Bounds.denseLipschitzBound;Purelib.Polynomial.Bounds.normSq_eval_sub_eval_le_denseLipschitzBound`.

Lemma 2.9 (finite Taylor remainder bound)**.** *Let $p \in \mathbb{Q}(i)[Z]$ and let B be a rational box. There is a displayed finite polynomial $T_p(z, w)$ satisfying*

$$p(z) - p(w) - p'(w)(z - w) = (z - w)^2 T_p(z, w), \tag{9}$$

and the same finite coefficient calculation computes $L_2(p, B) > 0$ such that, whenever $z, w \in B$,

$$\mathrm{N}(p(z) - p(w) - p'(w)(z - w)) \le L_2(p, B)\,\mathrm{N}(z - w)^2\,. \tag{10}$$

Proof. Expand every monomial $z^j = (w + (z - w))^j$ by the finite binomial identity and remove its constant and linear terms. Every remaining term contains $(z - w)^2$, giving the displayed T_p. Apply the same finite Horner bound as in lemma 2.8 and use $\mathrm{N}((z - w)^2) = \mathrm{N}(z - w)^2$. □

Formal verification:

`Purelib.Polynomial.Bounds.denseTaylorRemainderBound;Purelib.Polynomial.Bounds.normSq_dense_taylor_remainder_le_denseTaylorRemainderBound`.

The first estimate is the squared-distance form of the usual Lipschitz bound, retained so that every quantity remains rational. Both results are finite polynomial identities followed by rational inequalities.

The identity $Uf + Vf' = 1$ is the useful content of the squarefree certificate. Direct substitution says that f and f' cannot both vanish at one displayed argument; finite bounds on U, V strengthen this to the numerical rule needed below: a small value of f forces a non-small derivative. This is the local derivative control used by the extraction step.

Lemma 2.10 (finite squarefree-cofactor bounds)**.** *Let the squarefree certificate supply displayed $U, V \in \mathbb{Q}(i)[Z]$ satisfying the identity*

$$Uf + Vf' = 1.$$

For any rational box B, apply the Horner recurrence of lemma 2.4 separately to the high-to-low coefficient lists of U, V, obtaining H_U, H_V, and put

$$M_U = \max\{1, H_U\}, \qquad M_V = \max\{1, H_V\}.$$

Then $M_U, M_V \ge 1$, and every displayed $z \in B$ satisfies

$$\mathrm{N}(U(z)) \le M_U, \qquad \mathrm{N}(V(z)) \le M_V.$$

Proof. The two applications of lemma 2.4 give $\mathrm{N}(U(z)) \le H_U$ and $\mathrm{N}(V(z)) \le H_V$. Each output is at most its maximum with 1, which also proves the two positivity bounds. □

Formal verification:

`Purelib.Polynomial.Bounds.denseHornerNormBounds;Purelib.Polynomial.Bounds.normSq_eval_le_denseEvalNormBound`.

From these bounds define

$$\tau = \frac{1}{16 M_U}, \qquad \eta = \frac{1}{16 M_V}. \tag{11}$$

Lemma 2.11 (derivative margin from the Bézout certificate)**.** *For every rational z satisfying the predicate of B,*

$$\mathrm{N}(f(z)) \le \tau \quad \Longrightarrow \quad \mathrm{N}\big(f'(z)\big) \ge \eta.$$

Proof. From $1 = U(z)f(z) + V(z)f'(z)$ and (4),

$$1 \le 2M_U\,\mathrm{N}(f(z)) + 2M_V\,\mathrm{N}\big(f'(z)\big)\,.$$

If the displayed conclusion failed, then both products $M_U\,\mathrm{N}(f(z))$ and $M_V\,\mathrm{N}(f'(z))$ would be at most $1/16$. The right side would therefore be at most $1/4$, contradicting $1 \le 1/4$. □

Formal verification:

`Purelib.RootIsolation.Isolation.LocalExtraction.derivativeMargin;Purelib.RootIsolation.Isolation.LocalExtraction.derivativeMargin_le_of_small_residual;Purelib.RootIsolation.Isolation.LocalExtraction.derivative_ne_zero_of_small_residual`.

3 Sturm chains as finite programs

This section constructs the signed remainder chains whose endpoint variations give Cauchy indices. These indices are the inputs to box winding.

We use the signed Euclidean remainder sequence as the *definition* of the index over $\mathbb{Q}$. The proof below represents every crossing event by a rational isolating interval and a refinement routine. Throughout this paper, every Cauchy–Sturm index is normalized at twice the conventional scale: an interior crossing contributes ± 2 and an endpoint crossing contributes ± 1. This normalization is built directly into the endpoint variation defined below, avoiding an intermediate halving followed by a doubling in the index. We write simply Var_a and Ind_a^b, since no competing undoubled variation or index is used.

Definition 3.1 (rational interval partition)**.** *A rational interval $I = [a, b]$ is the endpoint pair (a, b) together with the predicate $a \le t \le b$ on a displayed rational t. An* interval partition *rooted at I is a finite binary tree of such endpoint records. An internal node $J = [u, v]$ stores a displayed rational cut c with $u < c < v$ and has the two children $[u, c]$ and $[c, v]$. A* leaf *is a terminal node, and the finitely many leaves of the tree are extracted as a finite list.*

Formal verification:

`Purelib.RootIsolation.Sturm.EventPartition.Cell;Purelib.RootIsolation.Sturm.EventPartition.SafeSplit;Purelib.RootIsolation.Sturm.EventPartition.Subdivision;Purelib.RootIsolation.Sturm.EventPartition.Subdivision.leaves`.

Definition 3.2 (central-third cut)**.** *For a proper rational interval $J = [u, v]$, its* open central third *is specified by the two rational bounds*

$$u + \frac{v-u}{3} < c < u + 2\frac{v-u}{3}.$$

A central-third cut *is a displayed rational c satisfying these inequalities.*

Lemma 3.3 (central-third contraction)**.** *If c is a central-third cut of $J = [u, v]$, then its two child widths obey*

$$c - u < \frac{2}{3}(v - u), \qquad v - c < \frac{2}{3}(v - u).$$

Proof. Subtract the left central-third inequality from c, and subtract the right one from v. □

Lemma 3.4 (finite safe-cut search)**.** *Let $H \in \mathbb{Q}[T]$ be a displayed nonzero polynomial, let $D \in \mathbb{N}$ be a displayed bound with $\deg H \le D$, and let J be a proper rational interval. A bounded search returns an H-safe central-third cut c: c lies in the open central third of J and $H(c) \neq 0$.*

Proof. Place $D + 1$ distinct, equally spaced rational candidates strictly between the two central-third bounds and evaluate H on that finite list. If all evaluations vanished, repeated exact division by the corresponding linear factors $T - c$ would force H to have degree at least $D + 1$. Hence one candidate has nonzero evaluation. Return the first such candidate in the displayed list. □

Formal verification:

`Purelib.RootIsolation.Isolation.SafeCandidates.centralLower;Purelib.RootIsolation.Isolation.SafeCandidates.centralUpper;Purelib.RootIsolation.Isolation.SafeCandidates.centralBounds;Purelib.RootIsolation.Isolation.SafeCandidates.centralSearchValue;Purelib.RootIsolation.Isolation.SafeCandidates.centralSearchValue_spec;Purelib.RootIsolation.Isolation.SafeCandidates.centralSearchValue_contraction`.

The central-third condition and a finite guard polynomial solve two independent problems at once. Choosing near the middle guarantees visible geometric progress, while avoiding the finitely many guarded zeros makes the new endpoint safe for the sign calculation encoded by H. The scan of $D + 1$ candidates is the finite substitute for saying "choose a generic cut." For a signed chain, the product of its members will be exactly such a guard.

3.1 Signs, variations, and signed remainders

Definition 3.5 (doubled sign variation)**.** *For $r \in \mathbb{Q}$, $\operatorname{sgn}(r) \in \{-1, 0, 1\}$ is decidable. For a finite list $s = (s_0, \ldots, s_n)$ define*

$$\operatorname{Var}(s) = \sum_{j=1}^{n} |\operatorname{sgn}(s_{j-1}) - \operatorname{sgn}(s_j)|. \tag{12}$$

This doubled variation is an integer: a direct switch between positive and negative contributes two, while a comparison between zero and a nonzero value contributes one. For polynomials $S_0, \ldots, S_n \in \mathbb{Q}[T]$ and $a \in \mathbb{Q}$, write

$$\operatorname{Var}_a(S_0, \ldots, S_n) = \operatorname{Var}(S_0(a), \ldots, S_n(a)), \qquad \operatorname{Var}_a^b = \operatorname{Var}_a - \operatorname{Var}_b\,.$$

Note this is twice the normal definition, but removing the $1/2$ here removes the need to double the contributions for the Cauchy–Sturm index calculation.

Formal verification:

`Purelib.RootIsolation.Sturm.signJump;Purelib.RootIsolation.Sturm.variation2;Purelib.RootIsolation.Sturm.endpointVariation2`.

To interpret this number, freeze the parameter at t and read the signs of the chain from left to right. Var_t records sign switches on the doubled scale fixed at the start of this section: a zero at one end of a comparison is a half-switch on the conventional scale and therefore contributes one here. As t moves, the signed remainder identities force changes at an interior chain member to cancel between its two neighbours. A net change survives only at the first member S_0, namely the denominator of R/S after GCD reduction. Thus $\mathrm{Var}_a - \mathrm{Var}_b$ is a finite algebraic ledger of the oriented pole-crossing events of R/S between the two endpoints. It extracts from endpoint signs the information ordinarily read from the graph of the quotient.

Definition 3.6 (reduced signed remainder chain)**.** *Given $R, S \in \mathbb{Q}[T]$ with $S \neq 0$, run the finite extended Euclidean program to remove their displayed common GCD g, and put $S_0 = S/g$. If $R = 0$, the returned chain is $[S_0]$. Otherwise put $S_1 = R/g$ and repeatedly perform*

$$S_{j+1} = Q_j S_j - S_{j-1}, \qquad \deg S_{j+1} < \deg S_j, \tag{13}$$

stopping immediately before the zero remainder. The resulting finite list of nonzero polynomials is the reduced signed remainder chain *of (R, S).*

Lemma 3.7 (termination and coprimality of the signed chain)**.** *The computation in definition 3.6 terminates after finitely many steps. Every consecutive pair in the returned chain is coprime, and hence no two consecutive members can vanish at the same displayed rational parameter.*

Proof. At every nonterminal Euclidean step, the natural degree count of the next remainder is strictly smaller than that of the current divisor. Recursion on that displayed natural number therefore terminates. The initial reduced pair comes with the Bézout identity returned by extended Euclid, and the identity in (13) transports coprimality to each consecutive pair. Indeed, if consecutive members S_j, S_{j+1} have displayed cofactors U, V with

$$US_j + VS_{j+1} = 1,$$

then substitution of any displayed $t \in \mathbb{Q}$ preserves this polynomial identity and gives

$$U(t)S_j(t) + V(t)S_{j+1}(t) = 1.$$

If both chain members vanished at t, the left-hand side would be zero, contradicting $0 \neq 1$ in $\mathbb{Q}$. □

Formal verification:

`Purelib.RootIsolation.Sturm.signedRemainderChain;Purelib.RootIsolation.Sturm.signedRemainderChain_length_le;Purelib.RootIsolation.Sturm.signedRemainderChain_all_nonzero;Purelib.RootIsolation.Sturm.noAdjacentZero_signedRemainderChain;Purelib.RootIsolation.Sturm.RootCount.reducedChain`.

Corollary 3.8 (safe cut for a signed remainder chain)**.** *For a reduced signed remainder chain $S_0, \ldots, S_m$, put*

$$G := \prod_{j=0}^{m} S_j.$$

There is a finite central-third search returning a cut c at which every $S_j(c)$ is nonzero. Both child intervals contract by the factor in lemma 3.3.

Proof. By definition, all members of the returned chain are nonzero polynomials, so their finite product G is nonzero. Apply lemma 3.4 to G and its computed degree. Since a nonzero product has no zero factor, $G(c) \neq 0$ proves $S_j(c) \neq 0$ for every member of the finite chain. The contraction conclusion is lemma 3.3. □

Formal verification:

`Purelib.RootIsolation.Sturm.RootCount.chainGuard;Purelib.RootIsolation.Sturm.RootCount.safeCut;Purelib.RootIsolation.Sturm.RootCount.safeCut_avoids_chain;Purelib.RootIsolation.Sturm.RootCount.safeCut_contraction`.

Definition 3.9 (finite Cauchy–Sturm index)**.** *For rational endpoints a, b, set the index to zero when $R = 0$ or $S = 0$. Otherwise, for the chain $S_0, \dots, S_m$ of definition 3.6, define*

$$\operatorname{Ind}_a^b\left(\frac{R}{S}\right) = \operatorname{Var}_a^b(S_0, S_1, \dots, S_m).$$

This is exactly the integer stored by the finite program on rational coefficients and endpoints. The doubled normalization is already built into (12), so no additional factor is inserted here. Reversing the endpoints changes the sign by definition.

Formal verification:

`Purelib.RootIsolation.Sturm.cauchyIndex2`.

The word *index* therefore refers to a signed event count, not to a location. A nonzero index certifies that the interval contains an uncancelled crossing event. Its sign records orientation, and its magnitude is the net count retained by the refinement machinery.

The local event scanner used below is entirely finite. It refines a rational interval to a coefficient-computed natural depth, records stable signs for the complete reduced chain on every leaf, and checks the resulting endpoint variation there.

Theorem 3.10 (finite Sturm calculation)**.** *For the displayed R, S of definition 3.9, the index and the safe partition program of corollary 3.8 satisfy the following identities.*

(a) Subdivision and reversal:

$$\operatorname{Ind}_a^b(R/S) + \operatorname{Ind}_b^c(R/S) = \operatorname{Ind}_a^c(R/S), \qquad \operatorname{Ind}_b^a(R/S) = -\operatorname{Ind}_a^b(R/S).$$

(b) If the displayed denominator is nonzero at both endpoints, the reduced index is even.

(c) If $S \neq 0$, then for an input rational interval $I = [a, b]$ a coefficient-computed finite event partition returns a finite leaf list. For each leaf $J = [u_J, v_J]$, its classification stores formal signs for the GCD-reduced numerator and denominator, proves that at least one of those signs is nonzero, proves their stability against the displayed endpoint evaluations at u_J and v_J, and identifies $\operatorname{Ind}_{u_J}^{v_J}(R/S)$ with the resulting finite endpoint event formula. The partition also proves the exact telescoping identity

$$\operatorname{Ind}_a^b(R/S) = \sum_{J \text{ a leaf}} \operatorname{Ind}_{u_J}^{v_J}(R/S).$$

Proof. Part (a) follows by addition and subtraction of the three displayed endpoint variations. For (b), the finite parity lemma for a coprime signed remainder chain is applied

at the two clear endpoints. For (c), use the safe central-third cut of corollary 3.8 at every internal node. A coefficient-computed depth makes the finite sign classifier succeed on every terminal interval. Its stored stability equalities give the endpoint event formula for that leaf. Subdivision distributes the parent index to its two children, and induction over the same finite depth telescopes the leaf list. □

Formal verification:

`Purelib.RootIsolation.Sturm.cauchyIndex2_add;Purelib.RootIsolation.Sturm.cauchyIndex2_reverse;Purelib.RootIsolation.Sturm.RootCount.reducedCauchyIndex2_even;Purelib.RootIsolation.Sturm.EventPartition.classifyEvents;Purelib.RootIsolation.Sturm.EventPartition.leafEvent_index2;Purelib.RootIsolation.Sturm.EventPartition.eventSubdivision_cellIndex2_sum`.

3.2 Univariate approximation tracks and positive lower bounds

Definition 3.11 (one-dimensional nonzero-event track)**.** *A nonzero-event track for a displayed nonzero polynomial p is a rational interval $[u, v]$ carrying the explicit certificate*

$$\mathrm{Ind}_u^v(p'/p) \neq 0,$$

together with the deterministic refinement program from theorem 3.10. A positive rational width request is answered by one finite nested interval of at most that width, again carrying an explicit nonzero-index certificate.

Formal verification:

`Purelib.RootIsolation.Sturm.OneDimensional.ZeroEventTrack`.

The retained certificate is exactly the nonzero signed event that survives each refinement request; it is not silently strengthened to count one.

Termination uses only the natural geometric-depth computation of lemma 2.7. Each safe central-third cut of corollary 3.8 contracts the retained interval by the fixed rational factor 2/3; the depth needed to cross a positive requested width is computed before the finite recursion starts.

Theorem 3.12 (finite one-dimensional event machinery)**.** *For rational polynomials R, S with $S \neq 0$ and a rational input interval $I = [a, b]$, the finite event-tree algorithm of theorem 3.10 classifies all leaves $J = [u_J, v_J]$ and proves*

$$\mathrm{Ind}_a^b(R/S) = \sum_{J \text{ a leaf}} \mathrm{Ind}_{u_J}^{v_J}(R/S).$$

For every explicitly selected nonzero-event track of definition 3.11, a total refinement function answers each positive rational width request by a nested interval of at most that width while retaining a nonzero Sturm index. Successive answers are nested by composition of their displayed rational refinement certificates.

Proof. Let V_t denote the doubled endpoint variation of the GCD-reduced signed chain at t. At an internal interval $[u, v]$, the safe cut c of corollary 3.8 gives the exact integer

calculation

$$\begin{aligned}\mathrm{Ind}_u^v(R/S) &= V_u - V_v \\ &= (V_u - V_c) + (V_c - V_v) \\ &= \mathrm{Ind}_u^c(R/S) + \mathrm{Ind}_c^v(R/S).\end{aligned}$$

Induction over the finite binary tree now replaces each parent by the sum of its two children. Write $\mathcal{I}([r,s]) = \mathrm{Ind}_r^s(R/S)$, and let L and M be the finite leaf lists of the left and right subtrees. Associativity of integer addition and the finite-list identity

$$\sum_{J \in L{+}{+}M} \mathcal{I}(J) = \sum_{J \in L} \mathcal{I}(J) + \sum_{J \in M} \mathcal{I}(J)$$

give exactly the displayed leaf sum. This is the subdivision part of theorem 3.10 written out algebraically.

Now let a track for p have current interval $[u,v]$ and nonzero count

$$C = \mathrm{Ind}_u^v(p'/p) \neq 0.$$

At its safe cut c, compute the two displayed integers

$$C_L = \mathrm{Ind}_u^c(p'/p), \qquad C_R = \mathrm{Ind}_c^v(p'/p), \qquad C = C_L + C_R.$$

Integer equality is decidable. If $C_L \neq 0$, retain $[u,c]$. Otherwise $C_L = 0$, and $C_R = 0$ would force $C = 0$; hence retain $[c,v]$ with the certificate $C_R \neq 0$. Thus one finite step preserves the nonzero count and refines the current interval.

If w_n is the retained width after n steps, the central-third inequalities give

$$w_{n+1} < \frac{2}{3} w_n, \qquad w_N \leq \left(\frac{2}{3}\right)^N w_0.$$

For a request $\varepsilon > 0$, the natural number $N = \mathrm{geometricDepth}(w_0, \varepsilon)$ of lemma 2.7 makes the last quantity at most ε. Executing exactly those N steps therefore returns the requested nonzero-event interval. For successive requests, the coordinate inequalities for the second returned interval refine those for the first, and those for the first refine the original; transitivity of the two endpoint inequalities proves the claimed nesting. □

Formal verification:

`Purelib.RootIsolation.Sturm.OneDimensional.classify;Purelib.RootIsolation.Sturm.OneDimensional.ClassifiedPartition.indexSum;Purelib.RootIsolation.Sturm.NonzeroEvent.step;Purelib.RootIsolation.Sturm.NonzeroEvent.refineN_refines;Purelib.RootIsolation.Sturm.NonzeroEvent.refineN_width_le;Purelib.RootIsolation.Isolation.SafeCandidates.geometricDepth_sufficient;Purelib.RootIsolation.Sturm.OneDimensional.ZeroEventTrack.request;Purelib.RootIsolation.Sturm.OneDimensional.ZeroEventTrack.request_refines;Purelib.RootIsolation.Sturm.OneDimensional.ZeroEventTrack.request_width_le;Purelib.RootIsolation.Sturm.OneDimensional.ZeroEventTrack.request_countNonzero;Purelib.RootIsolation.Sturm.OneDimensional.ZeroEventTrack.successive_refines`.

The next construction converts symbolic coprimality into a numerical safety margin. The identity $UP + VQ = 1$ says that P and Q cannot vanish simultaneously. Bounding the displayed cofactors U, V on the interval says more: the pair $(P(t), Q(t))$ cannot even become simultaneously too small. This is the recurring mechanism by which a finite algebraic certificate becomes boundary clearance.

Lemma 3.13 (finite Bézout clearance)**.** *Let $P, Q \in \mathbb{Q}[T]$, let $I = [a, b]$, and supply displayed U, V with the polynomial identity $UP + VQ = 1$. Using definition 2.5, put*

$$M = \max\{1, \mathcal{H}_I(U) + \mathcal{H}_I(V)\}, \qquad m = \frac{1}{M}.$$

Then $m \in \mathbb{Q}_{>0}$ and

$$P(t)^2 + Q(t)^2 \geq m$$

for every rational parameter satisfying $a \leq t \leq b$. The same certificate restricts unchanged to every displayed rational subinterval.

Proof. By lemma 2.6,

$$U(t)^2 + V(t)^2 \leq M.$$

The exact two-square identity gives

$$1 = (U(t)P(t) + V(t)Q(t))^2 \leq (U(t)^2 + V(t)^2)(P(t)^2 + Q(t)^2) \leq M(P(t)^2 + Q(t)^2).$$

Since $M \geq 1$, division gives $m = 1/M > 0$ and the required lower bound. A displayed rational subinterval has no new members, so the same inequality and the same m restrict unchanged. □

Formal verification:

`Purelib.Polynomial.Geometry.PathClearance.bezoutSqMajorant;Purelib.Polynomial.Geometry.PathClearance.certificateOfCoprime;Purelib.Polynomial.Geometry.PathClearance.Certificate.restrict`.

4 Algebraic winding on rational paths

This section combines the four Cauchy indices on a rational-box boundary into box winding and proves the finite rules used by subdivision and the rational-function action principle.

Definition 4.1 (oriented affine edges and box boundary)**.** *An oriented segment from u to v is the affine expression $e(t) = u + t(v - u)$ under the parameter predicate $0 \leq t \leq 1$. The boundary of a rational box B is its positively oriented four-element list* edges(B) *of such expressions. The notation ∂B used in prose abbreviates this finite list and its four parameter predicates.*

Clearance is the numerical guardrail around this boundary. Its positive margin says not only that the boundary value avoids zero, but by how much; that margin later permits perturbation, subdivision, and comparison without changing the winding calculation.

The intended winding picture is familiar: traverse the four edges and watch the direction of the polynomial or quotient value. Its net signed number of turns about zero is recovered here without angles or a continuous image curve, using only finitely many signed real-axis crossings.

Definition 4.2 (division-free quotient edge)**.** *For displayed $P, Q \in \mathbb{Q}(i)[Z]$ and an oriented edge e of definition 4.1, write*

$$\begin{aligned} P(e(t)) &= A_{P,e}(t) + iB_{P,e}(t), \\ Q(e(t)) &= A_{Q,e}(t) + iB_{Q,e}(t), \end{aligned} \qquad A_{P,e}, B_{P,e}, A_{Q,e}, B_{Q,e} \in \mathbb{Q}[t].$$

The division-free direction polynomial on the edge is

$$\begin{aligned} H_{P/Q,e}(t) &:= P(e(t))\overline{Q(e(t))} \\ &= A_{P,e}A_{Q,e} + B_{P,e}B_{Q,e} + i(B_{P,e}A_{Q,e} - A_{P,e}B_{Q,e}) \\ &=: C_{P/Q,e}(t) + iD_{P/Q,e}(t). \end{aligned}$$

Here conjugation is performed on the two rational coordinate polynomials of the affine edge; it is not treated as a holomorphic polynomial operation in Z.

Lemma 4.3 (division-free quotient direction)**.** *At a displayed edge parameter t, put $z = e(t)$. If $Q(z) \neq 0$, then*

$$\frac{P(z)}{Q(z)} = \frac{H_{P/Q,e}(t)}{\mathrm{N}(Q(z))}, \qquad \mathrm{N}(Q(z)) > 0. \tag{14}$$

If also $P(z) \neq 0$, then $H_{P/Q,e}(t) \neq 0$, so $C_{P/Q,e}(t)$ and $D_{P/Q,e}(t)$ do not vanish simultaneously.

Proof. The Gaussian-rational identities are

$$Q(z)\overline{Q(z)} = \mathrm{N}(Q(z))\,, \qquad Q(z)^{-1} = \frac{\overline{Q(z)}}{\mathrm{N}(Q(z))}.$$

The norm is a sum of two rational squares and is strictly positive when $Q(z) \neq 0$, proving (14). If $P(z)$ and $Q(z)$ are both nonzero, their product $P(z)\overline{Q(z)}$ is nonzero in the Gaussian-rational field; therefore its two rational coordinates cannot both be zero. □

Formal verification:

`Purelib.RationalFunctions.Winding.quotientPathReal;Purelib.RationalFunctions.Winding.quotientPathImag;Purelib.RationalFunctions.Winding.quotientEdge`.

Definition 4.4 (quotient winding)**.** *Let B carry boundary-clearance certificates for P and Q. The contribution of an oriented edge and the box winding are the finite Cauchy–Sturm calculations*

$$\mathrm{Wind}_4(P/Q; e) := \mathrm{Ind}_0^1\left(\frac{C_{P/Q,e}}{D_{P/Q,e}}\right), \qquad \mathrm{Wind}_4(P/Q; B) := \sum_{e \text{ in } \mathrm{edges}(B)} \mathrm{Wind}_4(P/Q; e),$$

where the index is that of definition 3.9. This formula uses only the displayed rational coordinate polynomials; it never evaluates a division by Q. The clearance certificates required in definition 1.11, together with lemma 4.3, prove that $H_{P/Q,e}$ never has both coordinates zero under an edge predicate.

Formal verification:

`Purelib.RationalFunctions.Winding.quotientBoundary;Purelib.RationalFunctions.Winding.rationalQuadrupledWinding`.

Concretely, along one edge the finite Sturm program reads the endpoint signs of $(C_{P/Q,e}, D_{P/Q,e})$ on rational event intervals. A change recorded in the sign ledger for $D_{P/Q,e}$ is the division-free surrogate for a real-axis crossing; the stored sign data for $C_{P/Q,e}$ and the orientation of that change determine its signed contribution. The program

preserves this integer event count together with its rational interval and sign certificates. When the four oriented edges are joined, endpoint half-events at a shared vertex cancel, and the remaining signed crossing total is exactly the net number of turns. This is the bridge between the sign variations of the preceding section and the box winding used throughout the rest of the paper.

Lemma 4.5 (polynomial specialization)**.** *For $P = f$ and $Q = 1$, the quotient-edge construction satisfies $H_{f/1,e} = f(e) = A_{f,e} + iB_{f,e}$ and hence*

$$\mathrm{Wind}_4(f; e) = \mathrm{Ind}_0^1\left(\frac{A_{f,e}}{B_{f,e}}\right), \qquad \mathrm{Wind}_4(f; B) = \sum_{e \text{ in } \mathrm{edges}(B)} \mathrm{Wind}_4(f; e).$$

This is the polynomial specialization of definition 4.4.

Proof. The coordinate polynomials of the constant denominator 1 are 1 and 0. Hence definition 4.2 gives

$$\begin{aligned} H_{f/1,e} &= f(e)\overline{1} = f(e), \\ C_{f/1,e} &= A_{f,e} \cdot 1 + B_{f,e} \cdot 0 = A_{f,e}, \\ D_{f/1,e} &= B_{f,e} \cdot 1 - A_{f,e} \cdot 0 = B_{f,e}. \end{aligned}$$

Unfolding the edge definition therefore gives

$$\mathrm{Wind}_4(f; e) = \mathrm{Wind}_4(f/1; e) = \mathrm{Ind}_0^1\left(\frac{C_{f/1,e}}{D_{f/1,e}}\right) = \mathrm{Ind}_0^1\left(\frac{A_{f,e}}{B_{f,e}}\right).$$

Summing this equality over the four entries of $\mathrm{edges}(B)$ proves the box identity. □

Formal verification:

`Purelib.RationalFunctions.Winding.rationalQuadrupledWinding;Purelib.RootIsolation.Winding.quadrupledBoxWinding`.

Every Wind_4 value is obtained by four finite Euclidean chains, and every winding-one state stores the exact equality $\mathrm{Wind}_4(f; B) = 4$.

Lemma 4.6 (edge reversal)**.** *For an oriented affine edge e, put $e^{\mathrm{rev}}(t) := e(1-t)$. Whenever the edge winding of a displayed polynomial or division-free quotient expression F is defined,*

$$\mathrm{Wind}_4(F; e^{\mathrm{rev}}) = -\mathrm{Wind}_4(F; e).$$

Proof. The coordinate polynomials on e^{rev} are those on e after the affine change $t \mapsto 1-t$. Thus traversing e^{rev} from $t = 0$ to $t = 1$ is the same finite Cauchy–Sturm calculation as traversing e from $t = 1$ to $t = 0$. The reversal identity in theorem 3.10 negates that index. □

Formal verification:

`Purelib.RootIsolation.Sturm.cauchyIndex2_reverse`.

Lemma 4.7 (box subdivision)**.** *If a vertical or horizontal rational cut divides a proper box B into B_1 and B_2, then*

$$\mathrm{Wind}_4(f; B) = \mathrm{Wind}_4(f; B_1) + \mathrm{Wind}_4(f; B_2).$$

The corresponding statement holds for a finite rectangular grid.

Proof. In the sum of the two child boundary lists, the internal cut occurs once in each orientation. Its two contributions cancel by lemma 4.6. Each outer edge of B is either unchanged or divided into two consecutive rationally parametrized segments; the subdivision identity in theorem 3.10 replaces the contributions of those segments by the contribution of the original edge. The remaining sum is therefore exactly $\mathrm{Wind}_4(f; B)$. Iterating the same finite argument over the cuts proves the rectangular-grid case. □

Formal verification:

`Purelib.RootIsolation.Winding.quadrupledBoxWinding_vertical_split;Purelib.RootIsolation.Winding.quadrupledBoxWinding_horizontal_split`.

Proposition 4.8 (coordinate winding)**.** *For every displayed $R > 0$, let $B = [-R, R]^2$. For the coordinate polynomial Z, one has $\mathrm{Wind}_4(Z; B) = 4$.*

Proof. Substitute the four affine edge parametrizations of the symmetric square. Each index is the variation of a chain of degree at most one; the finite sign table gives edge contributions $0, 2, 0, 2$, whose sum is four. □

Formal verification:

`Purelib.RootIsolation.Winding.Global.coordinate_symmetric_normalization`.

Proposition 4.9 (winding parity)**.** *If a displayed polynomial f carries a boundary-clearance certificate on a proper rational box B, then $\mathrm{Wind}_4(f; B)$ is even.*

Proof. The four endpoint parity corrections telescope at the shared vertices of the oriented boundary, so the box sum is even. □

Formal verification:

`Purelib.RootIsolation.Winding.quadrupledBoxWinding_even_of_boundaryCoprime`.

Proposition 4.10 (product formula)**.** *If f and g carry quantitative boundary-clearance certificates on a proper rational box B, then*

$$\mathrm{Wind}_4(fg; B) = \mathrm{Wind}_4(f; B) + \mathrm{Wind}_4(g; B).$$

In particular, for every nonzero constant c, $\mathrm{Wind}_4(c; B) = 0$.

Proof. On one edge write $f = A + iB$ and $g = C + iD$. Then

$$\mathrm{Re}(fg) = AC - BD, \qquad \mathrm{Im}(fg) = AD + BC.$$

The finite local product calculation has one correction determined by the endpoint signs of (A, B, C, D). Its generic and coordinate-degenerate branches are finite sign tables. Summing around the four sides cancels each outgoing correction with the incoming correction at the same vertex; the clearance hypotheses remove simultaneous-zero cases. What remains is the displayed identity. This follows the algebraic product strategy of Eisermann [5], but does not invoke its unrestricted auxiliary formula. Perrucci and Roy [7] explain why that formula needs additional cases. Here a finite guard partition resolves simultaneous coordinate events, the coordinate-degenerate branches are checked separately, and only then are the endpoint corrections telescoped. □

Formal verification:

`Purelib.RootIsolation.Winding.Product.quadrupledBoxWinding_mul_of_boundaryClearances`.

Theorem 4.11 (finite quotient winding formula)**.** *Let $P, Q \in \mathbb{Q}(i)[Z]$, and let C be a proper rational box whose boundary carries positive clearance certificates for both polynomials. The generic division-free construction of definition 4.4 satisfies*

$$\mathrm{Wind}_4(P/Q; C) = \mathrm{Wind}_4(P; C) - \mathrm{Wind}_4(Q; C).$$

Proof. On one oriented affine edge write

$$P(e) = A + iB, \qquad Q(e) = C + iD.$$

The direction polynomial from the generic construction is

$$P(e)\overline{Q(e)} = (AC + BD) + i(BC - AD).$$

Apply the finite product calculation underlying proposition 4.10 to the coordinate pairs (A, B) and $(C, -D)$. It expresses the edge index of their product as the sum of their two edge indices plus a correction determined entirely by the endpoint signs. Conjugation changes the sign of the second edge index. After the four oriented edges are added, the outgoing correction at each vertex cancels the incoming correction at that same vertex. Boundary clearance excludes the simultaneous-zero cases in every finite sign table. The remaining identity is therefore the stated subtraction of windings. Every step is a signed Euclidean-remainder calculation over $\mathbb{Q}$. □

Formal verification:

`Purelib.RationalFunctions.Winding.rationalQuadrupledWinding_eq_sub_of_boundaryClearances`.

4.1 Certified homotopies and a finite Rouché lemma

Definition 4.12 (dominance certificate)**.** *A* dominance certificate *for displayed polynomial expressions F, G on a rationally parametrized boundary consists of a positive rational margin m and a finite polynomial-inequality proof of*

$$\mathrm{N}(F(t)) - \mathrm{N}(G(t)) \geq m > 0$$

under each rational edge predicate.

Every later use constructs the margin in definition 4.12 directly from finite coefficient bounds; no unbounded search is hidden in the certificate. Intuitively, it records enough room to add G to F without allowing the boundary value to reach zero.

Lemma 4.13 (half-plane winding)**.** *If the four coordinate-polynomial expressions obtained by composing a displayed polynomial h with the edges of a proper rational box B have quantitative clearance and satisfy $\mathrm{Re}\, h(e(t)) > 0$ under their rational parameter predicates, then $\mathrm{Wind}_4(h; B) = 0$.*

Proof. The imaginary part may cross zero, but the real part is positive at every such crossing. In the endpoint variation formula the two half-crossings on adjacent edges cancel, and no crossing of the negative real ray occurs. Equivalently, apply the product formula after multiplying by a fixed rational constant; the four composed edge expressions then carry certificates for the displayed half-plane predicate. □

Formal verification:

`Purelib.Polynomial.Geometry.HalfPlane.quadrupledBoxWinding_eq_zero_of_positive_real`.

Lemma 4.14 (finite dominance homotopy)**.** *Let F, G be displayed polynomial expressions on a proper rational-box boundary, with clearance for F. A dominance certificate in the sense of definition 4.12 proves*

$$\mathrm{Wind}_4(F+G;B) = \mathrm{Wind}_4(F;B).$$

Proof. On each affine edge form the displayed coordinate polynomial for $\overline{F}(F+G)$. The exact two-square identity and the supplied positive margin make its real coordinate positive throughout the edge. Finite Horner bounds turn the same margin into a clearance certificate for $F+G$. The half-plane formula and the finite product formula then identify the edge index of $F+G$ with that of F, up to one explicit endpoint correction. The four corrections telescope around the box, leaving the displayed winding identity. □

Formal verification:

`Purelib.RootIsolation.Winding.Homotopy.quadrupledBoxWinding_add_of_boundaryNormDominance`.

Corollary 4.15 (finite Rouché rule)**.** *Whenever a rational norm bound proves $\mathrm{N}(g(z)) < \mathrm{N}(f(z))$ on the clear boundary of a proper rational box B,*

$$\mathrm{Wind}_4(f+g;B) = \mathrm{Wind}_4(f;B).$$

Proof. The supplied strict rational inequality is a dominance certificate. Lemma 4.14 gives the displayed identity. □

Formal verification:

`Purelib.RootIsolation.Winding.Homotopy.quadrupledBoxWinding_add_of_boundaryNormDominance`.

5 Clearing axis-parallel lines

This section constructs eliminants and uses them to compute safe vertical and horizontal cuts. Their boundary-clearance certificates make rational-box subdivision available to local extraction.

Write

$$f(x+iy) = A_f(x,y) + iB_f(x,y), \qquad A_f, B_f \in \mathbb{Q}[x,y].$$

A vertical coordinate $x = c$ is usable when an explicit Bézout calculation rules out simultaneous vanishing of the two displayed coordinate polynomials along that line; a

horizontal coordinate is treated analogously. The two eliminants below are univariate guard polynomials for those calculations. A nonzero guard value at c lets us construct the required Bézout and quantitative-clearance certificates using finite arithmetic. The guards thereby choose safe grid lines.

5.1 Safe axis-parallel cuts

Definition 5.1 (fraction-free scaled-inverse recursion)**.** *Let M be a square matrix over $\mathbb{Q}(i)[T]$. Its scaled-inverse data (s, N) are defined by recursion on its dimension. In dimension zero put $s = 1$ and take the empty matrix for N. If M has positive dimension $n+1$, write*

$$M = \begin{pmatrix} a & u \\ v & M_{\text{tail}} \end{pmatrix}, \qquad S = aM_{\text{tail}} - vu,$$

where a is a scalar, u is a $1 \times n$ row, v is an $n \times 1$ column, and M_{tail} and S are $n \times n$ matrices. If recursion on the smaller matrix S returns (s', N'), return

$$s = as', \qquad N = \begin{pmatrix} s' + uN'v & -auN' \\ -aN'v & a^2N' \end{pmatrix}.$$

Lemma 5.2 (scaled-inverse identity)**.** *If the recursion of definition 5.1 returns (s, N) from M, then the following identities hold over $\mathbb{Q}(i)[T]$:*

$$NM = sI = MN.$$

Proof. Induct on the dimension. The zero-dimensional case is immediate. In positive dimension, for $S = aM_{\text{tail}} - vu$, the induction hypothesis gives the polynomial-matrix identities

$$N'S = s'I = SN'.$$

Multiplying the two block matrices in definition 5.1, the off-diagonal blocks of NM reduce to

$$s'u - uN'(aM_{\text{tail}} - vu) = s'u - us' = 0, \qquad -a^2N'v + a^2N'v = 0,$$

while its diagonal blocks reduce to as' and $aN'S = as'I$. Thus $NM = as'I = sI$. Multiplication in the other order uses $SN' = s'I$ and gives $MN = as'I$ in the same way. □

Formal verification:

`Purelib.Polynomial.SeparatedVariables.scaledInverseData_coefficientIdentity`.

Definition 5.3 (separated-variable Schur–Sylvester eliminants)**.** *Put*

$$P_T(U) = f(U), \qquad Q_T(U) = \overline{f}(2T - U),$$

where $\overline{f}$ means coefficientwise conjugation, and let $S_x(T)$ be their displayed $2d \times 2d$ Sylvester coefficient matrix over $\mathbb{Q}(i)[T]$. Write (s_x, N_x) for the scaled-inverse data of S_x from definition 5.1. Define $E_x \in \mathbb{Q}[T]$ by taking coefficientwise the imaginary part of s_x when the real part of its leading coefficient is zero, and otherwise taking its coefficientwise real part.

Apply the same construction to the quarter-turned polynomial $f^{\circlearrowleft}(Z) = f(iZ)$; write (s_y, N_y) for its scaled-inverse data and $E_y \in \mathbb{Q}[T]$ for the rational coordinate selected by the same rule. Finally put

$$D_f = \max\{\deg E_x, \deg E_y\}.$$

Lemma 5.4 (nonzero Schur–Sylvester eliminants)**.** *If f is nonconstant, then E_x and E_y of definition 5.3 are nonzero.*

Proof. Write a_d for the nonzero leading coefficient of f. In each of the first d columns of $S_x(T)$, coming from Q_T, use d as the displayed parameter-degree bound; in each of the last d columns, coming from $P_T = f$, use degree zero. Reading the finite coefficient lists shows that the matrix of these column-leading coefficients is upper triangular. Its first d diagonal entries are

$$2^d\overline{a_d},$$

the top parameter coefficient created by repeatedly substituting $2T - U$ in $\overline{f}$, and its last d diagonal entries are a_d. Both are nonzero in $\mathbb{Q}(i)$. Entries strictly below the diagonal have degree strictly below their column's displayed bound.

Now consider one Schur step of definition 5.1. Suppose the current matrix has column-degree bounds δ_j and nonzero diagonal leading coefficients λ_j. The head entry has leading term $\lambda_0 T^{\delta_0}$. In column j of

$$S = aM_{\text{tail}} - vu,$$

the first product has degree at most $\delta_0 + \delta_{j+1}$ and diagonal leading coefficient $\lambda_0\lambda_{j+1}$. In the correcting product vu, the entry from the left column lies strictly below the original diagonal, so its degree is too small to cancel that leading term. Thus S again has an upper-triangular leading-coefficient matrix, now with diagonal entries $\lambda_0\lambda_{j+1}$, all nonzero.

Induction therefore shows that the head entry at every recursive stage is a nonzero polynomial. Since the recursion starts with scalar 1 and replaces s' by as', its output s_x is a finite product of those nonzero pivots and is itself nonzero. Let $\alpha + i\beta$ be the leading coefficient of the normalized s_x. If $\alpha \neq 0$, the coefficientwise real part has leading coefficient α; if $\alpha = 0$, then $\beta \neq 0$ and the coefficientwise imaginary part has leading coefficient β. The rule in definition 5.3 therefore makes E_x nonzero. Finally, $f^{\circlearrowleft}(Z) = f(iZ)$ has the same degree and leading coefficient $a_d i^d \neq 0$, so the identical argument gives $E_y \neq 0$. □

Formal verification:

`Purelib.Polynomial.SeparatedVariables.separatedSylvester_leading;Purelib.LinearAlgebra.PolynomialMatrix.Leading.scaledInverseData_scalar_nonzero;Purelib.Polynomial.SeparatedVariables.separatedScalar_nonzero;RFAP.ConstructiveSylvesterBezout.scalarEliminant_nonzero;RFAP.ConstructiveSylvesterBezout.horizontalEliminant_nonzero;Purelib.RootIsolation.Isolation.AxisSafeCuts.separatedEliminationDegree.`

Lemma 5.5 (specialized axis Bézout certificates)**.** *For every displayed $c \in \mathbb{Q}$ with $E_x(c) \neq 0$, finite calculation returns $U_c, V_c \in \mathbb{Q}[Y]$ with the polynomial identity*

$$1 = U_c(Y)A_f(c, Y) + V_c(Y)B_f(c, Y).$$

Likewise, $E_y(c) \neq 0$ returns displayed polynomials satisfying the corresponding polynomial identity for $A_f(X, c)$ and $B_f(X, c)$.

Proof. Coefficientwise real and imaginary projection commute with Horner evaluation at the rational parameter c. Consequently $E_x(c)$ is either the real or the imaginary coordinate of $s_x(c)$. If $s_x(c)$ were zero, that selected coordinate would be zero as well; hence

$$E_x(c) \neq 0 \quad \Longrightarrow \quad s_x(c) \neq 0.$$

Use the right-hand identity of lemma 5.2 and take the first column after specializing at c:

$$S_x(c)N_x(c)e_0 = s_x(c)e_0.$$

Split the displayed column $N_x(c)e_0$ into its first and last d entries, say $(\ell_0, \dots, \ell_{d-1})$ and $(r_0, \dots, r_{d-1})$, and form

$$L_c(U) = \sum_{j<d} \ell_j U^j, \qquad R_c(U) = \sum_{j<d} r_j U^j.$$

Multiplying the Sylvester rows by $1, U, \dots, U^{2d-1}$ and summing is a finite rearrangement of coefficients. The first d columns reconstruct $Q_c(U)$ and the last d columns reconstruct $P_c(U)$, so the displayed matrix identity gives the polynomial identity

$$L_c(U)Q_c(U) + R_c(U)P_c(U) = s_x(c)$$

in $\mathbb{Q}(i)[U]$. Division by the now certified nonzero scalar $s_x(c)$ gives polynomials $\widehat{L}_c, \widehat{R}_c$ satisfying

$$\widehat{L}_c(U)Q_c(U) + \widehat{R}_c(U)P_c(U) = 1. \tag{15}$$

Substitute the polynomial $c+iY$ into (15). Since $\overline{2c - (c+iY)} = c+iY$, the definitions of the separated pair give

$$P_c(c+iY) = f(c+iY), \qquad Q_c(c+iY) = \overline{f(c+iY)}.$$

Write $f(c+iY) = A_f(c,Y) + iB_f(c,Y)$. Taking the real part of the substituted polynomial identity yields

$$\begin{aligned} 1 = {} & \big(\operatorname{Re}\widehat{L}_c(c+iY) + \operatorname{Re}\widehat{R}_c(c+iY)\big)A_f(c,Y) \\ & + \big(\operatorname{Im}\widehat{L}_c(c+iY) - \operatorname{Im}\widehat{R}_c(c+iY)\big)B_f(c,Y). \end{aligned}$$

The two parenthesized expressions are displayed polynomials $U_c, V_c \in \mathbb{Q}[Y]$, proving the vertical claim.

For the horizontal claim apply the same construction to $f^{\circlearrowleft}(w) = f(iw)$. A horizontal point $t + ic$ is obtained from the displayed parameter $w = c - it$ because $iw = t + ic$. The vertical identity for the quarter-turned polynomial therefore becomes the asserted identity for $A_f(X,c)$ and $B_f(X,c)$ after the same real-coordinate calculation. □

Formal verification:

`RFAP.ConstructiveSylvesterBezout.scalar_ne_zero_of_eliminant_eval_ne_zero; RFAP.ConstructiveSylvesterBezout.verticalCoprimeCertificate;RFAP.ConstructiveSylvesterBezout.horizontalCoprimeCertificate`.

Lemma 5.6 (bounded safe-cut search)**.** *Let $[a,b]$ be a nondegenerate rational x-interval and let $[r,s]$ be a bounded rational y-interval. Apply lemma 3.4 to $H = E_x$ with the degree bound D_f of definition 5.3. The resulting finite scan returns c in the open central third of $[a,b]$ with $E_x(c) \neq 0$ and computes a positive rational m_c such that*

$$A_f(c,t)^2 + B_f(c,t)^2 \geq m_c \qquad (r \leq t \leq s).$$

Thus the full vertical line segment $c+it$ carries a quantitative boundary clearance certificate. Interchanging x and y gives the analogous safe horizontal cut.

Proof. By lemma 5.4, E_x is nonzero and $\deg E_x \le D_f$. The bounded search in lemma 3.4 therefore returns the stated c after at most $D_f + 1$ exact evaluations.

Apply lemma 5.5 to $E_x(c) \neq 0$ to obtain the displayed polynomials U_c, V_c and their Bézout identity. Apply the finite Bézout-clearance calculation of lemma 3.13 to this identity on $[r, s]$. It returns $m_c > 0$ with $A_f(c,t)^2 + B_f(c,t)^2 \ge m_c$ for every displayed rational t satisfying $r \le t \le s$. This is exactly $m_c \le \mathrm{N}(f(c+it))$. The calculation for E_y is identical after interchanging the coordinates. □

Formal verification:

`Purelib.RootIsolation.Isolation.AxisSafeCuts.separatedAxisEliminationCertificate;Purelib.RootIsolation.Isolation.AxisSafeCuts.verticalSafeCoordinate_spec;Purelib.RootIsolation.Isolation.AxisSafeCuts.horizontalSafeCoordinate_spec;Purelib.RootIsolation.Isolation.AxisSafeCuts.verticalSafeCoordinate_contraction;Purelib.RootIsolation.Isolation.AxisSafeCuts.horizontalSafeCoordinate_contraction;Purelib.RootIsolation.Isolation.AxisSafeCuts.buildClearFourGrid`.

The inequalities in lemma 3.3 make both child widths strictly less than 2/3 of the parent width. This is the contraction used by every refinement routine.

6 The degree winding on a large square

This section computes the global winding budget on the coefficient isolation square from displayed coefficient bounds.

Let

$$f(Z) = a_d Z^d + \sum_{k=0}^{d-1} a_k Z^k, \qquad a_d \neq 0.$$

Recall from Definition 1.6 that

$$C_f = d \sum_{k=0}^{d-1} 2^k \,\mathrm{N}(a_k)$$

is the displayed weighted norm bound for the nonleading coefficients. Take the coefficient radius $R = R_f$ from the same definition. It satisfies

$$1 \le R, \qquad C_f < \mathrm{N}(a_d)\, R \le \mathrm{N}(a_d)\, R^2. \tag{16}$$

These are direct rational comparisons from the displayed formula (2); no search over a geometric object is involved. The square is called *global* only because every output track uses it as a common containing box. It remains finite endpoint data, and the displayed inequalities are the leading-term margin used below.

Definition 6.1 (leading monomial, lower tail, and dominance margin)**.** *For the displayed polynomial f and coefficient radius R, put*

$$L_f(Z) = a_d Z^d, \qquad T_f(Z) = \sum_{k=0}^{d-1} a_k Z^k, \qquad f = L_f + T_f.$$

The associated leading–tail dominance margin *is the displayed rational*

$$M_f = (\mathrm{N}(a_d)\, R^2 - C_f)(R^2)^{d-1}.$$

Lemma 6.2 (leading–tail annular margin)**.** *The margin M_f is positive. Every displayed $q \in \mathbb{Q}(i)$ satisfying*

$$R^2 \le \mathrm{N}(q) \le 2R^2$$

also satisfies

$$M_f \le \mathrm{N}(L_f(q)) - \mathrm{N}(T_f(q))\,.$$

Proof. Positivity follows from $C_f < \mathrm{N}(a_d)\,R^2$ in (16) and $R > 0$. For the annular estimate, (4), the upper bound on $\mathrm{N}(q)$, and $R \ge 1$ give

$$\begin{aligned}\mathrm{N}(T_f(q)) &\le d \sum_{k<d} \mathrm{N}(a_k)\,(2R^2)^k \\ &= d \sum_{k<d} 2^k\,\mathrm{N}(a_k)\,R^{2k} \\ &\le C_f R^{2d-2}.\end{aligned}$$

The lower bound on $\mathrm{N}(q)$ gives

$$\mathrm{N}(L_f(q)) = \mathrm{N}(a_d)\,\mathrm{N}(q)^d \ge \mathrm{N}(a_d)\,R^{2d}.$$

Subtracting the two rational bounds yields

$$\mathrm{N}(L_f(q)) - \mathrm{N}(T_f(q)) \ge \big(\mathrm{N}(a_d)\,R^2 - C_f\big)R^{2d-2} = M_f.$$

□

Formal verification:

`Purelib.RootIsolation.Winding.Global.leadingMonomial;Purelib.RootIsolation.Winding.Global.lowerTermsPolynomial;Purelib.RootIsolation.Winding.Global.leadingMonomial_add_lowerTerms_eval;Purelib.RootIsolation.Winding.Global.largeSquareDominanceMargin;Purelib.RootIsolation.Winding.Global.largeSquareDominanceMargin_pos;Purelib.RootIsolation.Winding.Global.largeSquare_radial_dominance_margin`.

Let $K_R = (-R, R, -R, R)$. Its boundary satisfies $R^2 \le \mathrm{N}(z) \le 2R^2$, so Lemma 6.2 gives

$$0 < M_f \le \mathrm{N}(L_f(z)) - \mathrm{N}(T_f(z))\,.$$

The meaning of this estimate is simple: on the outer square the leading monomial is larger than the sum of all lower terms, so the finite Rouché rule allows the tail to be discarded without changing winding. Multiplication by Z contributes one turn, and the product rule therefore makes Z^d contribute d. This is the sole source of the global count used later.

Theorem 6.3 (global degree winding)**.** *The boundary of K_R is clear and*

$$\mathrm{Wind}_4(f; K_R) = 4d, \qquad \mathrm{Wind}_f(K_R) = d.$$

Proof. The dominance inequality of Lemma 6.2 and corollary 4.15 give

$$\mathrm{Wind}_4(f; K_R) = \mathrm{Wind}_4(L_f; K_R).$$

A nonzero constant has winding zero, Z has winding four by proposition 4.8, and repeated use of proposition 4.10 gives $\mathrm{Wind}_4(L_f; K_R) = 4d$. The conventional value $\mathrm{Wind}_f(K_R) = d$ follows from definition 1.11. □

Formal verification:

`Purelib.RootIsolation.Winding.Global.quadrupledBoxWinding_largeSquare_eq_degree`.

7 Local extraction from a nonzero-winding box

This section performs local extraction: from a boundary-clear box with nonzero box winding, it extracts one winding-one root state and retains a boundary-clear subdivision carrying all the remaining winding. Applied to a winding-one state, the same construction is the refinement step later packaged as a RAT.

More precisely, put $w = \mathrm{Wind}_4(f; B) \neq 0$. For any displayed request $\varepsilon > 0$, the algorithm returns four safe cut lines and their full clear 3×3 subdivision of B. Its central box D is strictly inside B and satisfies

$$\mathrm{Wind}_4(f; D) = 4, \qquad \mathrm{diag}^2(D) \leq \varepsilon^2.$$

If $\mathcal{R}$ is the list of the eight noncentral boxes, lemma 4.7 gives

$$w = 4 + \sum_{E \in \mathcal{R}} \mathrm{Wind}_4(f; E).$$

Thus only the selected box is normalized to winding four. If $w > 4$, total winding $w - 4$ remains in the complementary boxes; if $w = 4$, their total is zero. Before nonnegativity is proved, w may also be negative; the same equation then gives the complementary total $w - 4$. The fact that each extraction removes exactly four from the retained remainder is the step iterated in the counting argument of section 9.

The construction has five stages.

(i) Safely subdivide B until a nonzero-winding child C is small and lies away from the parent's boundary.
(ii) Use the small diameter and nonzero winding to prove that the rational anchor a_C has a small residual $\mathrm{N}(f(a_C))$.
(iii) Use the squarefree Bézout certificate to prove that $f'(a_C)$ is not small, and hence compute a rational Newton center.
(iv) Use the explicit Taylor bound to show that the affine part about that center dominates the nonlinear remainder on a small surrounding band.
(v) Choose four eliminant-safe lines in that band. Their central rectangle is the returned box D; the full grid records both D and its eight complementary boxes.

The constants introduced below merely allocate enough rational margin for these five steps to hold simultaneously.

7.1 Finite local bounds and the protected boundary

Definition 7.1 (local finite Horner bounds)**.** *Fix a current rational box B and write*

$$R_B = 2 \max\{x_-^2, x_+^2, y_-^2, y_+^2\}.$$

For the displayed high-to-low coefficient list $[a_n, \dots, a_0]$ *of* f*, start with* $(M_0, S_0, T_0) = (0, 0, 0)$ *and, for* $0 \le k \le n$*, compute*

$$\begin{aligned} M_{k+1} &= 2\big(M_k R_B + \mathrm{N}(a_{n-k})\big), \\ S_{k+1} &= 2\big(S_k R_B + M_k\big), \\ T_{k+1} &= 2\big(S_k + R_B T_k\big). \end{aligned}$$

Define the rationals

$$L_1(B) = \max\{1, S_{n+1}\}, \qquad L_2(B) = \max\{1, T_{n+1}\}.$$

Lemma 7.2 (local Horner estimates)**.** *The quantities of definition 7.1 are positive, since*

$$L_1(B), L_2(B) \ge 1.$$

For every pair of displayed Gaussian rationals z, w *satisfying the membership predicate of* B*, they also satisfy*

$$\mathrm{N}(f(z) - f(w)) \le L_1(B)\,\mathrm{N}(z - w)\,, \tag{17}$$

$$\mathrm{N}(f(z) - f(w) - f'(w)(z - w)) \le L_2(B)\,\mathrm{N}(z - w)^2\,. \tag{18}$$

Proof. Run Horner evaluation simultaneously on the displayed coefficient prefixes at z and w. At stage k, M_k bounds the two partial values, S_k bounds their divided difference, and T_k bounds the remainder after removing its constant and linear terms. Multiplication by the next argument uses $\mathrm{N}(z)\,, \mathrm{N}(w) \le R_B$; adjoining the next coefficient uses the finite two-term inequality $\mathrm{N}(x + y) \le 2(\mathrm{N}(x) + \mathrm{N}(y))$. These are exactly the three updates defining $M_{k+1}, S_{k+1}, T_{k+1}$. Induction over the finite coefficient list therefore gives the two displayed estimates with S_{n+1} and T_{n+1}. Replacing them by their maxima with 1 preserves the inequalities and gives $L_1(B), L_2(B) > 0$. □

Thus $L_1(B)$ is the squared-distance Lipschitz coefficient and $L_2(B)$ is the quadratic Taylor-remainder coefficient. They are obtained by one finite fold over the coefficient list; neither is a maximum or supremum over the points satisfying the box predicate. In the notation of (8) and (10), they are $L_1(f, B)$ and $L_2(f, B)$.

Formal verification:

`Purelib.Polynomial.Bounds.HornerNormBounds.step;Purelib.Polynomial.Bounds.denseHornerNormBounds;Purelib.Polynomial.Bounds.denseLipschitzBound;Purelib.Polynomial.Bounds.HornerTaylorNormBounds.step;Purelib.Polynomial.Bounds.denseHornerTaylorNormBounds;Purelib.Polynomial.Bounds.denseTaylorRemainderBound;Purelib.Polynomial.Bounds.normSq_eval_sub_eval_le_denseLipschitzBound;Purelib.Polynomial.Bounds.normSq_dense_taylor_remainder_le_denseTaylorRemainderBound`.

We continue to write $\tau(B), \eta(B)$ for the quantities in (11) computed on B, and $\mu(B) > 0$ for the bound in its four-edge clearance certificate.

Definition 7.3 (protected boundary radius)**.** *Define*

$$\rho(B) = \min\left\{\frac{1}{512}, \frac{\mu(B)}{2L_1(B)}, \frac{\mathrm{width}_x(B)}{16}, \frac{\mathrm{width}_y(B)}{16}\right\}$$

from the displayed clearance, Horner bound, and endpoint data of B*.*

Lemma 7.4 (positivity of the protected radius)**.** *If B is proper and carries the stated clearance certificate, then $\rho(B) > 0$.*

Proof. Properness of B makes both coordinate widths positive. The clearance certificate gives $\mu(B) > 0$, while lemma 7.2 gives $L_1(B) > 0$. Hence all four entries in the finite minimum of definition 7.3 are positive. □

Formal verification:

`Purelib.Polynomial.Geometry.ProtectedBoundary.protectedBoundaryRadius;`
`Purelib.Polynomial.Geometry.ProtectedBoundary.protectedBoundaryRadius_pos`.

Lemma 7.5 (protected boundary strip)**.** *Let B be proper and carry the stated clearance certificate. If a rational subbox $D \subseteq B$ has $\operatorname{diag}^2(D) \le \rho(B)^2$ and meets the coordinate boundary strip of width $4\rho(B)$, then $\operatorname{Wind}_4(f; D) = 0$.*

Proof. Put $\rho = \rho(B)$, $L = L_1(B)$, and $\mu = \mu(B)$. Properness of B makes both coordinate widths positive; L, μ, and ρ are positive by their finite certificates and lemma 7.4.

Meeting the strip supplies a displayed $z \in D$ and one of four rational side inequalities. Consider the left-side case

$$\operatorname{Re} z \le x_-^B + 4\rho;$$

the other three cases are the same calculation with the relevant coordinate reversed. Since $D \subseteq B$, also $x_-^B \le \operatorname{Re} z$. Project z to the displayed parent-boundary point

$$b = x_-^B + i \operatorname{Im} z.$$

Then b satisfies the left-edge parametrization predicate of B and

$$\mathrm{N}(b - z) = (\operatorname{Re} z - x_-^B)^2 \le 16\rho^2.$$

Let $q = a_D$ be the fixed lower-left anchor of D. Membership of z and q in D gives

$$\mathrm{N}(z - q) \le \operatorname{diag}^2(D) \le \rho^2.$$

Using $b - q = (b - z) + (z - q)$ and the finite two-term quadrance inequality,

$$\begin{aligned} \mathrm{N}(b - q) &\le 2\big(\mathrm{N}(b - z) + \mathrm{N}(z - q)\big) \\ &\le 2(16\rho^2 + \rho^2) = 34\rho^2 \le 64\rho^2. \end{aligned}$$

The point b satisfies the boundary-membership predicate of B. Also $q \in D$ and $D \subseteq B$, so q satisfies the membership predicate of B. We may therefore apply (17) to the displayed pair (b, q). Substituting the preceding distance estimate gives

$$\mathrm{N}(f(b) - f(q)) \le L\,\mathrm{N}(b - q) \le L(64\rho^2) = 64L\rho^2,$$

and hence

$$\mathrm{N}(f(b) - f(q)) \le 64L\rho^2. \tag{19}$$

Set

$$A = L \operatorname{diag}^2(D), \qquad V = 64L\rho^2.$$

The definition of ρ gives

$$A \le L\rho^2, \qquad \rho \le \frac{1}{512}, \qquad 2L\rho \le \mu.$$

Since $130/512 < 2$ and $L, \rho > 0$, these rational inequalities yield

$$2(A + V) \le 130L\rho^2 < 2L\rho \le \mu. \tag{20}$$

The boundary-clearance certificate at b, followed by the two-term quadrance inequality and (19), gives

$$\mu \le \mathrm{N}(f(b)) \le 2\big(\mathrm{N}(f(q)) + \mathrm{N}(f(b) - f(q))\big) \le 2\big(\mathrm{N}(f(q)) + V\big).$$

If $\mathrm{N}(f(q)) \le A$, this contradicts (20), because the preceding display would then give the impossible chain

$$\mu \le 2(A + V) < \mu.$$

Consequently

$$A < \mathrm{N}(f(q)). \tag{21}$$

Finally, let x be any displayed point on one of the four edge parametrizations of D. Both x and q satisfy the membership predicate of D, and hence, because $D \subseteq B$, that of B. We may apply (17) once more; the box-diagonal certificate then gives

$$\mathrm{N}(f(x) - f(q)) \le L\,\mathrm{N}(x - q) \le L\,\mathrm{diag}^2(D) = A < \mathrm{N}(f(q)).$$

Moreover $A \ge 0$, so (21) implies $f(q) \neq 0$. In the exact polynomial decomposition

$$f = f(q) + (f - f(q)),$$

the nonzero constant polynomial $f(q)$ therefore strictly dominates the second summand on the complete rational boundary of D. Lemma 4.14 identifies $\mathrm{Wind}_4(f; D)$ with the winding of that constant, which is zero by proposition 4.10. Hence $\mathrm{Wind}_4(f; D) = 0$, as required. □

Formal verification:

`Purelib.Polynomial.Geometry.ProtectedBoundary.nearbyBoundaryPoint;Purelib.Polynomial.Geometry.ProtectedBoundary.protectedBoundaryRadius_dominates;Purelib.Polynomial.Geometry.ProtectedBoundary.quadrupledBoxWinding_eq_zero_of_boundaryStripWitness;Purelib.Polynomial.Geometry.ProtectedBoundary.strictly_inside_protectedBoundary;Purelib.RootIsolation.Isolation.LocalExtraction.quadrupledBoxWinding_eq_zero_of_anchorResidualBound_lt`.

7.2 Newton center and extraction

Lemma 7.6 (small nonzero box gives a rational Newton center)**.** *Let $C \subseteq B$ be a rational box with $\mathrm{Wind}_4(f; C) \neq 0$, fixed lower-left anchor $a = a_C \in \mathbb{Q}(i)$, and squared diagonal δ^2. Then*

$$\mathrm{N}(f(a)) \le L_1(B)\delta^2.$$

If in addition $L_1(B)\delta^2 \le \tau(B)$, then $\mathrm{N}(f'(a)) \ge \eta(B)$, and consequently

$$c = a - \frac{f(a)}{f'(a)} \in \mathbb{Q}(i), \qquad \mathrm{N}(c - a) \le \frac{L_1(B)\delta^2}{\eta(B)}.$$

Proof. Every displayed boundary point z of C lies with a in B, and $\mathrm{N}(z-a) \le \delta^2$. Hence lemma 7.2 gives

$$\mathrm{N}(f(z) - f(a)) \le L_1(B)\delta^2.$$

If the claimed residual bound failed, the nonzero constant $f(a)$ would strictly dominate $f - f(a)$ on all four edges of C. The finite Rouché rule would then give $\mathrm{Wind}_4(f; C) = 0$, contrary to the supplied certificate. Therefore $\mathrm{N}(f(a)) \le L_1(B)\delta^2$.

Under the additional hypothesis this is at most $\tau(B)$, so lemma 2.11 gives $\mathrm{N}(f'(a)) \ge \eta(B) > 0$. In particular $f'(a) \neq 0$, and field arithmetic constructs the displayed $c \in \mathbb{Q}(i)$. Finally,

$$\mathrm{N}(c-a) = \mathrm{N}\left(\frac{f(a)}{f'(a)}\right) = \frac{\mathrm{N}(f(a))}{\mathrm{N}(f'(a))} \le \frac{L_1(B)\delta^2}{\eta(B)},$$

using multiplicativity of the Gaussian-rational norm and the two preceding bounds. □

Formal verification:

`Purelib.RootIsolation.Isolation.LocalExtraction.anchor_residual_le_of_quadrupledBoxWinding_ne_zero;Purelib.RootIsolation.Isolation.LocalExtraction.computeNewtonCenterOfRawWinding`.

Definition 7.7 (local extraction scales)**.** *For a proper cleared box B and a request $\varepsilon > 0$, define*

$$r_{\mathrm{ext}}(B, \varepsilon) = \min\left\{\frac{\rho(B)}{2}, 1, \frac{\eta(B)}{648L_2(B)}, \frac{\varepsilon}{8}\right\},$$

and, abbreviating this radius to r, define the source-diagonal threshold

$$\theta_{\mathrm{src}}(B, \varepsilon) = \min\left\{\rho(B)^2, \frac{\tau(B)}{L_1(B)}, \frac{\eta(B)r^2}{L_1(B)}\right\}.$$

Lemma 7.8 (extraction-scale budgets)**.** *Let B and $\varepsilon > 0$ satisfy the hypotheses of definition 7.7. Put $r = r_{\mathrm{ext}}(B, \varepsilon)$ and $\theta = \theta_{\mathrm{src}}(B, \varepsilon)$. Then $r, \theta > 0$ and*

$$2r \le \rho(B), \qquad r \le 1, \qquad 648L_2(B)r \le \eta(B), \qquad 8r \le \varepsilon,$$
$$\theta \le \rho(B)^2, \qquad L_1(B)\theta \le \tau(B), \qquad L_1(B)\theta \le \eta(B)r^2.$$

Proof. The quantities $\rho(B), \tau(B), \eta(B), L_1(B), L_2(B)$ and ε are positive, so every entry in both displayed finite minima is positive. The four bounds on r are the four minimum projections, with positive denominators cleared. The three bounds on θ follow identically from its three minimum projections. □

Formal verification:

`Purelib.RootIsolation.Newton.Extraction.extractionRadius;Purelib.RootIsolation.Newton.Extraction.extractionRadius_pos;Purelib.RootIsolation.Newton.Extraction.extractionRadius_twice_le_protected;Purelib.RootIsolation.Newton.Extraction.extractionRadius_le_one;Purelib.RootIsolation.Newton.Extraction.extractionRadius_le_eighth;Purelib.RootIsolation.Newton.Extraction.extractionRadius_taylor;Purelib.RootIsolation.Newton.Extraction.sourceDiagonalThreshold;Purelib.RootIsolation.Newton.Extraction.sourceDiagonalThreshold_pos;Purelib.RootIsolation.Newton.Extraction.sourceDiagonalThreshold_le_protected;`

`Purelib.RootIsolation.Newton.Extraction.sourceDiagonalThreshold_le_derivative;Purelib.RootIsolation.Newton.Extraction.sourceDiagonalThreshold_le_displacement`.

Definition 7.9 (Newton band)**.** *Let $c \in \mathbb{Q}(i)$ and $r \in \mathbb{Q}_{>0}$. A* Newton band *of centre c and radius r is a rational box D satisfying*

$$\begin{aligned} \operatorname{Re} c - 2r \le x_-^D \le \operatorname{Re} c - r, \quad \operatorname{Re} c + r \le x_+^D \le \operatorname{Re} c + 2r, \\ \operatorname{Im} c - 2r \le y_-^D \le \operatorname{Im} c - r, \quad \operatorname{Im} c + r \le y_+^D \le \operatorname{Im} c + 2r. \end{aligned}$$

Thus c lies strictly inside D, both side lengths are at most $4r$, and $\operatorname{diag}^2(D) \le 32r^2$. An eliminant-safe Newton band *additionally carries the four full-line clearance certificates obtained by selecting its sides with the bounded safe-cut searches.*

Formal verification:

`Purelib.RootIsolation.Newton.Band.NewtonBandBox;Purelib.RootIsolation.Newton.Band.NewtonBandBox.center_strictly_inside;Purelib.RootIsolation.Newton.Band.NewtonBandBox.diagonalSq_le`.

Theorem 7.10 (local extraction)**.** *Fix the displayed squarefree and axis-elimination certificates. Let B be a proper rational box carrying certificates for ambient containment, boundary clearance, and $\operatorname{Wind}_4(f; B) \neq 0$. There is a terminating algorithm*

$$\begin{aligned} &\operatorname{extractOne}(f, B, \mathcal{W}, \varepsilon), \\ &\mathcal{W} : \operatorname{Wind}_4(f; B) \neq 0, \qquad \varepsilon \in \mathbb{Q}_{>0}. \end{aligned}$$

which returns rational coordinates

$$x_- < x_+, \qquad y_- < y_+,$$

strictly inside B such that:

(a) the four full axis-parallel cut lines through these coordinates are safe within B;
(b) the central box $D = (x_-, x_+, y_-, y_+)$ has $\operatorname{Wind}_4(f; D) = 4$;
(c) splitting B along the four lines gives a finite 3×3 grid with clear boundaries, the central member being D;
(d) $\operatorname{diag}^2(D) \le \varepsilon^2$.

Proof. Put $\rho = \rho(B)$, $r = r_{\mathrm{ext}}(B, \varepsilon)$, and $\theta = \theta_{\mathrm{src}}(B, \varepsilon)$. Their positivity and all required rational budgets are given by lemma 7.8. At each subdivision round, apply the safe-cut construction of lemma 3.4 to the current x-coordinate interval with $H = E_x$, and to the y-coordinate interval with $H = E_y$. Pairing the two x-interval children with the two y-interval children gives four rational child boxes, each of whose side widths is less than $2/3$ of the corresponding parent width. Run this finite four-child recursion to its precomputed geometric depth and retain the first nonzero-winding child C with $\operatorname{diag}^2(C) \le \theta$. By lemma 7.5, C lies in the protected core of B.

Let $a = a_C$ be its anchor and let

$$c = a - \frac{f(a)}{f'(a)}$$

be the rational Newton center from lemma 7.6. The definition of θ and lemmas 7.6 and 7.8 give $\mathrm{N}(c - a) \le r^2$, while the protected-core inequalities and $2r \le \rho$ keep the square of half-side $2r$ about c strictly inside B.

For z in that square,

$$\mathrm{N}(z-c) \le 8r^2, \qquad \mathrm{N}(z-a) \le 2\big(\mathrm{N}(z-c)+\mathrm{N}(c-a)\big) \le 18r^2.$$

Put

$$\ell(z) = f(a) + f'(a)(z-a) = f'(a)(z-c).$$

On any candidate rectangular boundary surrounding c whose coordinate distance from c is at least r and at most $2r$,

$$\mathrm{N}(\ell(z)) = \mathrm{N}\big(f'(a)\big)\,\mathrm{N}(z-c) \ge \eta(B)r^2.$$

The exact Taylor identity and the two corresponding radius budgets in lemma 7.8 give the finite dominance estimate

$$\mathrm{N}(f(z)-\ell(z)) \le 324L_2(B)r^4 \le \frac{1}{2}\eta(B)r^2.$$

Thus, for every such rectangular boundary enclosing c, the quantitative Rouché rule and propositions 4.8 and 4.10 give $\mathrm{Wind}_4(f;E) = \mathrm{Wind}_4(\ell;E) = 4$ for its enclosed box E.

Finally run the bounded eliminant search of lemma 5.6 in each of the four rational bands

$$[\mathrm{Re}\,c-2r, \mathrm{Re}\,c-r], \quad [\mathrm{Re}\,c+r, \mathrm{Re}\,c+2r], \quad [\mathrm{Im}\,c-2r, \mathrm{Im}\,c-r], \quad [\mathrm{Im}\,c+r, \mathrm{Im}\,c+2r].$$

The selected full cut lines are safe, and their central rectangle D surrounds c while staying in the affine-dominance square. Hence $\mathrm{Wind}_4(f;D) = 4$, and the four lines produce the certified 3×3 grid. Both side lengths are at most $4r$, so $\mathrm{diag}^2(D) \le 32r^2 \le \varepsilon^2$. □

Formal verification:

`Purelib.RootIsolation.Newton.Extraction.extract`.

8 Finite derivative–Taylor track separation

This section turns coefficient bounds and squarefree certificates into separation certificates for RATs. Separation is the input needed later for finite maximality.

The local extractor can be called inside any nonzero-winding box. To bound the length of every finite family of separated outputs in a bounded square, we first compute a scale on which the derivative term cannot be cancelled by both a quadratic Taylor error and two sufficiently small displayed residuals. This is the same coefficient machinery already used by local extraction.

Every observation considered in this section is a RAT, so its support record in Definition 1.16 is a proper rational box. Later refinements move the relevant endpoints inward, preserving the same displayed inequality and gap. This witnessed finite relation is what *distinct* means below.

The underlying idea is uniqueness of a good linear approximation. If two separated tracks both fitted inside one sufficiently small box, their rational anchors would be distinct by a witnessed gap, yet both polynomial values could be made arbitrarily small. Taylor's identity would then say that the non-small derivative times the anchor difference is simultaneously large enough to detect that gap and small enough to be absorbed by the two residuals and the quadratic error. The explicit scale v_f is where those two conclusions become incompatible.

Lemma 8.1 (finite uniqueness and residual budgets)**.** *On the coefficient isolation square K_R, put*

$$\upsilon_f = \frac{\eta(K_R)}{8L_2(f, K_R)} > 0.$$

For every displayed coordinate gap $h > 0$, the rational quantities

$$\beta_h = \min\left\{\tau(K_R), \frac{\eta(K_R)h^2}{32}\right\} > 0, \qquad r_h = \min\left\{1, \frac{\beta_h}{L_1(f, K_R)}\right\} > 0$$

are computable. Calling a winding-one RAT with request r_h returns an anchor u satisfying

$$\mathrm{N}(f(u)) \le \beta_h.$$

Proof. Positivity follows by finite rational comparison from the positive squarefree derivative margin and the positive Horner bounds. The returned box has squared diagonal at most $r_h^2 \le r_h$. Applying the anchor residual estimate and the definition of r_h gives $L_1(f, K_R)r_h^2 \le \beta_h$. □

Formal verification:

`Purelib.RootIsolation.Isolation.TrackSeparation.uniquenessDiagonalThreshold;Purelib.RootIsolation.Isolation.TrackSeparation.uniquenessDiagonalThreshold_pos;Purelib.RootIsolation.Isolation.TrackSeparation.pairResidualThreshold;Purelib.RootIsolation.Isolation.TrackSeparation.pairResidualThreshold_pos;Purelib.RootIsolation.Isolation.TrackSeparation.residualRefinementRadius;Purelib.RootIsolation.Isolation.TrackSeparation.refined_anchor_residual_le`.

Theorem 8.2 (root-track separation)**.** *The positive rational υ_f of lemma 8.1 has the following finite property. Two separated winding-one RATs cannot both have current boxes contained in one rational box $C \subseteq K_R$ with* $\mathrm{diag}^2(C) < \upsilon_f$.

Proof. Suppose two separated tracks have current boxes in such a C. Their separation certificate gives a positive rational coordinate gap h. Request the residual radius r_h of lemma 8.1 from both tracks, and call the returned anchors u, v. Put $t = v - u$. Separation and common-box membership give

$$0 < h^2 \le \mathrm{N}(t) \le \mathrm{diag}^2(C) < \upsilon_f.$$

The two residual requests give

$$\mathrm{N}(f(u)), \mathrm{N}(f(v)) \le \beta_h,$$

while the derivative margin gives

$$\eta(K_R) \le \mathrm{N}(f'(u)).$$

Write the exact finite Taylor identity as

$$f(v) - f(u) = f'(u)t + E, \qquad \mathrm{N}(E) \le L_2(f, K_R)\,\mathrm{N}(t)^2.$$

The inequality $\mathrm{N}(t) < \upsilon_f$ bounds $8\,\mathrm{N}(E)$ by $\eta(K_R)\,\mathrm{N}(t)$. The definition of β_h, the two residual bounds, and $h^2 \le \mathrm{N}(t)$ likewise bound $8\,\mathrm{N}(f(v) - f(u))$ by $\eta(K_R)\,\mathrm{N}(t)$. Two applications of the finite two-term quadrance inequality would then force

$$4\eta(K_R)\,\mathrm{N}(t) \le 2\eta(K_R)\,\mathrm{N}(t),$$

contradicting the displayed positivity of both factors. Thus the common box cannot exist. □

Formal verification:

`Purelib.RootIsolation.Isolation.TrackSeparation.separated_tracks_not_supported_in_small_common_box`.

Corollary 8.3 (finite packing bound)**.** *There is a computable natural number P_f such that no finite list of pairwise separated winding-one RATs supported in K_R has length greater than P_f.*

Proof. Choose a rational mesh $s > 0$ with $8s^2 < v_f$ and subdivide the endpoint record for $[-R, R]^2$ into an explicit finite grid list of half-open cell records of side at most s. Ask every member of a proposed finite list for a box of coordinate width below $s/4$, and assign it to the cell whose membership predicate accepts its rational anchor. If two anchors receive the same cell, both refined boxes lie in the coordinatewise enlargement of that cell by $s/4$; this common rational box has squared diagonal below v_f, contradicting theorem 8.2. Finite pigeonhole induction therefore bounds the list length by the number of cells; take that number as P_f. □

Formal verification:

`Purelib.RootIsolation.Isolation.Packing.packingBound;Purelib.RootIsolation.Isolation.Packing.family_card_le_packingBound`.

9 Nonnegativity and exact extraction of d tracks

This section combines the global winding budget, local extraction, and separation to construct exactly d RATs, proves ownership and finite maximality, and relates small residuals to local and global refinement.

At this point Wind_4 is an additive integer, and every nonzero-winding box contains a finitely extractable winding-one subbox. The separation theorem forces the remaining sign.

The argument is a finite counting trap. If a clear box had negative winding, extracting a winding-one box from any nonzero remainder and removing its contribution would leave the remainder negative, so the operation could be repeated any prescribed finite number of times. Repeating it more than the finite packing bound would produce too many separated winding-one boxes. Negative winding is therefore impossible. Once every local contribution is nonnegative, the global total $4d$ can be spent only as exactly d contributions of four; this is the accounting step that produces the initial tracks.

Theorem 9.1 (nonnegativity)**.** *Every clear rational box $B \subseteq K_R$ satisfies*

$$\mathrm{Wind}_4(f; B) \geq 0.$$

The statement also holds for a finite disjoint list of clear boxes, with total winding defined by finite summation.

Proof. Assume a finite list of rational-box records has negative total winding. Choose a member with nonzero winding and apply theorem 7.10 with request 1. Remove the central

winding-one box and replace the old member by the other eight grid boxes. Additivity decreases the total by exactly four, so it remains negative and the operation may be repeated.

Perform this finite loop exactly $P_f + 1$ times. It returns $P_f + 1$ pairwise distinct winding-one boxes, each carrying the deterministic refinement routine of theorem 7.10. This contradicts the finite packing bound corollary 8.3. Therefore the original total could not be negative. □

Formal verification:

`Purelib.RootIsolation.Isolation.GlobalNonnegativity.quadrupledBoxWinding_nonnegative`.

Theorem 9.2 (exact finite isolation)**.** *Starting from K_R, a recursion on d returns exactly d pairwise separated clear rational boxes $B_1, \dots, B_d$ with*

$$\mathrm{Wind}_4(f; B_j) = 4.$$

Every remaining grid box has winding zero.

Proof. The initial total is $4d$ by theorem 6.3. If the current total is positive, some member has nonzero winding; apply theorem 7.10 with request 1 and replace its parent by the eight complements. By additivity the remaining total decreases by four. By theorem 9.1 it never becomes negative. After exactly d iterations the remaining total is zero, and nonnegativity forces every remaining member to have winding zero. □

Formal verification:

`Purelib.RootIsolation.Isolation.ExactIsolation.isolateLargeSquare`.

Operationally, a RAT is a certified zoom command: supply its current finite record and a requested scale, and it returns one smaller certified record. The bounded search may inspect several candidate boxes internally, but one request has one observable answer. The word *track* denotes the ability to repeat this interaction. The zero/one/many distinction belongs to the enclosing isolation algorithm, which returns an exact terminal or a finite list of tracks.

Lemma 9.3 (total strict winding-one refinement)**.** *Let B be a rational box carrying a certified winding-one state, with the displayed squarefree and axis-elimination certificates. For every $\varepsilon \in \mathbb{Q}_{>0}$ the local Newton-band extractor returns a rational box D carrying the next certified state and satisfying*

$$D \subset B, \qquad \mathrm{Wind}_4(f; D) = 4, \qquad \mathrm{diag}^2(D) \le \varepsilon^2.$$

The same finite construction, stored as the state transition, can be called again on D.

Proof. Apply theorem 7.10 to the current state and the requested ε. Its returned four line-clearance certificates, Wind_4 calculation, strict-containment inequalities, and diagonal estimate are exactly the fields of the next finite state. Reapplying the same definition therefore gives a total request-driven refinement track. □

Formal verification:

`Purelib.RootIsolation.Isolation.ShrinkingTrack.refineState_accurate`.

Output ownership. Fix the output of theorem 9.2. Its partition has output boxes $B_1, \dots, B_d$ and zero-winding remainder cells. Let $m_\Pi > 0$ be the minimum of the finitely many displayed cell-clearance bounds, and define

$$\rho_{\mathrm{own}} := min\left\{1, \frac{m_\Pi}{8L_1(f, K_f)}\right\} > 0.$$

For a competing RAT T supported in the same coefficient isolation square, make the single call at this radius and put

$$D_T := \mathrm{box}(T(\rho_{\mathrm{own}})).$$

An *ownership certificate* for T is an index $j \in \{1, \dots, d\}$ and a proof of strict containment

$$D_T \subset B_j.$$

The deterministic selector scans the finite partition for the anchor of D_T and proves that the whole box, not merely its anchor, lies strictly inside the selected cell. A zero-winding remainder cell cannot be selected, so the scan returns an output index j. We write $\mathrm{own}(T) = j$ and call the jth returned RAT the *owner* of T.

Proof of the polynomial part of theorem 1.17. If $d = 1$, exact finite coefficient normalization and division compute the displayed Gaussian-rational zero and verify it by evaluation. Suppose $d \geq 2$. Use theorem 9.2 for the d initial states and lemma 9.3 for every accuracy call. Let q_B be the fixed rational anchor of a returned box. The nonzero-winding argument in lemma 7.6 gives

$$\mathrm{N}(f(q_B)) \leq L_1(f, A)\,\mathrm{diag}^2(B).$$

For an arbitrary displayed $z \in B$, equation (8) and the finite sum inequality give the uniform, deliberately looser estimate

$$\mathrm{N}(f(z)) \leq 2(\mathrm{N}(f(z) - f(q_B)) + \mathrm{N}(f(q_B))) \leq 4L_1(f, A)\,\mathrm{diag}^2(B).$$

Equation (5) proves the bound on Hermitian projective quadrance, and substitution of (6) gives the opposite-corner Hermitian-projective-quadrance–residual formula in theorem 1.17. If $q_{B'}$ is the anchor of a refining box, both anchors satisfy the old membership predicate, hence $\mathrm{N}(q_{B'} - q_B) \leq \mathrm{diag}^2(B)$.

The isolation routine leaves a finite safe rectangular partition consisting of the d output boxes and boxes with $\mathrm{Wind}_4 = 0$. Let an additional winding-one RAT be given. The positive lower bounds on the finitely many partition edges, together with the residual estimate just proved, give a finite refinement after which its box meets no partition edge; hence it lies in one partition member. It cannot lie in a zero-winding member: safe cuts around its smaller winding-one box would make additivity and nonnegativity give total winding at least four there. It therefore lies in an output box B_j.

If it were separated from the output RAT in B_j, refine the two to disjoint boxes with a positive rational gap. Safe-cut searches in that gap extend them to two disjoint members of a finite grid inside B_j. Nonnegativity and additivity would give $\mathrm{Wind}_4(f; B_j) \geq 8$, contradicting $\mathrm{Wind}_4(f; B_j) = 4$. Thus every pairwise separated family injects, by a finite cell assignment, into the d output boxes. Its length is at most d. □

Formal verification:

`Purelib.RootIsolation.Isolation.Maximality.family_card_le_natDegree;`
`Purelib.RootIsolation.Isolation.BoxFTA.provePositive`.

The main theorem gives the easy direction: a sufficiently small winding-one box makes the residual at its anchor small. The next result asks for a controlled converse. Starting from a displayed sample with small residual, can we build a certified nearby track? The answer must be local: we also need a protected box and its squarefree derivative margin, since the bare number $\mathrm{N}(f(q))$ changes when f is rescaled. With that context, one rational Newton step proposes a center and the same Taylor–Rouché calculation constructs the box around it.

Definition 9.4 (protected parent)**.** *For a displayed polynomial f and fixed ambient box A, a* protected parent *is a proper rational box B, a refinement $B \subseteq A$, and a boundary-clearance certificate for f on B. No winding value is part of this record. A winding-one state supplies a protected parent by forgetting its winding certificate.*

Formal verification:

`Purelib.RootIsolation.Winding.Residual.ProtectedAmbient;Purelib.RootIsolation.Winding.Residual.ProtectedAmbient.ofWindingOne`.

Theorem 9.5 (local residual-to-Hermitian-projective-quadrance converse)**.** *Let f have degree at least two and carry a squarefree certificate, let B be a rational box carrying a certified winding-one state, and let $q \in \mathbb{Q}(i)$ satisfy the strict interior predicate of B. For every $\varepsilon \in \mathbb{Q}_{>0}$ one can compute a positive rational $\Delta(f, B, q, \varepsilon)$ with the following property. If*

$$\mathrm{N}(f(q)) < \Delta(f, B, q, \varepsilon),$$

then exact evaluation either returns the finite certificate $f(q) = 0$, or a finite Newton–Rouché construction returns a rational box D and a RAT beginning at D such that

$$D \subset B, \qquad \mathrm{Wind}_4(f; D) = 4, \qquad \mathrm{diag}^2(D) \le 32\varepsilon^2, \qquad \chi^2(q, z) \le 18\varepsilon^2 \quad (z \in D).$$

The output exposes the selected positive rational radius $r \le \varepsilon$, the winding certificate 4, the four strict-containment inequalities, and the stronger displayed estimate

$$\mathrm{N}(z - q) \le 18r^2 \quad (z \in D).$$

If only ordinary membership $q \in B$ is initially known, a second computable positive target first uses one quarter of the certified boundary residual clearance to prove strict membership and then gives the same conclusion.

Proof. Let $C_2(f, B) > 0$ denote the explicit finite Taylor coefficient used by the selected safe Newton band. Compute r as the minimum of

$$1,\ \frac{\mathrm{Re}\, q - x_-}{6},\ \frac{x_+ - \mathrm{Re}\, q}{6},\ \frac{\mathrm{Im}\, q - y_-}{6},\ \frac{y_+ - \mathrm{Im}\, q}{6},\ \varepsilon,\ \frac{\tau(B)}{4L_1(f, B)},\ \frac{\eta(B)}{C_2(f, B)},$$

and put

$$\Delta(f, B, q, \varepsilon) = \min\{\, 2L_1(f, B)r^2,\ \eta(B)r^2 \,\}. \tag{22}$$

Every entry is a displayed positive rational, so r and Δ are positive. The Bézout identity and the first residual budget give $\mathrm{N}(f'(q)) \ge \eta(B) > 0$. Thus the rational Newton point

$$n = q - \frac{f(q)}{f'(q)}$$

is defined, and exact norm multiplicativity gives

$$\mathrm{N}(n-q) = \frac{\mathrm{N}(f(q))}{\mathrm{N}(f'(q))} \le r^2.$$

The four geometric entries keep the eliminant-safe band around n strictly inside B. The Taylor entry makes the nonlinear remainder subordinate to the affine factor $f'(q)(Z-n)$ on every selected side. Finite Rouché transfer therefore gives $\mathrm{Wind}_4(f;D) = 4$, and the standard refiner supplies the total RAT.

Every band point is within two coordinate radii of n, while the Newton displacement puts each coordinate of n within one radius of q. Hence both coordinate differences from q are at most $3r$, so

$$\mathrm{N}(z-q) \le (3r)^2 + (3r)^2 = 18r^2 \le 18\varepsilon^2.$$

Equation (5) gives the Hermitian-projective quadrance conclusion. For the ordinary-membership form, the minimum additionally includes one quarter of the finite boundary-clearance lower bound and replaces the four pointwise gaps by the computed uniform interior-strip radius. □

Formal verification:

`Purelib.Polynomial.Geometry.ResidualChordal.ofResidual;Purelib.Polynomial.Geometry.ResidualChordal.observeOfResidual`.

The preceding theorem accepts a requested radius ε and tests whether the residual is small enough for it. The next corollary runs that relation in the other numerical direction: starting from the residual R, it computes a rational radius whose square is within a chosen factor $\kappa > 1$ of R/μ. The factor κ is deliberate rational slack that replaces an exact square root; the finite dyadic bracket is not a new geometric object.

Corollary 9.6 (requested-factor rational residual bound)**.** *Under the hypotheses of theorem 9.5, put*

$$\mu(f,B) = \min\{\eta(B),\, 2L_1(f,B)\} > 0$$

and let $s = s(f,B,q) > 0$ be the reference radius obtained from the geometric, derivative-threshold, and Taylor caps in that proof with the requested cap $\varepsilon = 1$. For any displayed rational $\kappa > 1$, define

$$\Lambda_\kappa(f,B,q) = \min\left\{\mu(f,B), \frac{\mu(f,B)s^2}{\kappa}\right\} > 0.$$

If $0 < R = \mathrm{N}(f(q)) < \Lambda_\kappa(f,B,q)$, a bounded dyadic scan followed by a precomputed finite number of exact rational comparisons computes a positive rational ρ satisfying

$$\frac{R}{\mu(f,B)} < \rho^2 \le \frac{\kappa R}{\mu(f,B)}.$$

The resulting strictly nested winding-one box D and its RAT obey

$$\chi^2(q,z) \le \frac{18\kappa}{\mu(f,B)}R \quad (z \in D), \qquad \mathrm{diag}^2(D) \le \frac{32\kappa}{\mu(f,B)}R.$$

These estimates are local to the supplied winding-one state on B.

Proof. Write $\mu = \mu(f, B)$ and put $x = R/\mu$. Since $0 < R < \Lambda_\kappa \le \mu$, the displayed rational x lies strictly between zero and one. Starting at 1, repeatedly halve the rational radius. An explicit natural depth computed from x makes the final dyadic square no larger than x, so this is a bounded recursion rather than a square-root operation or an unbounded search. It returns the last radius r whose square is above x; one halving step gives

$$x < r^2 \le 4x.$$

Put

$$g = \min\left\{1, \frac{\kappa - 1}{3}\right\} > 0.$$

The rational inequalities $g \le 1$ and $3g \le \kappa - 1$ give

$$(1+g)^2 = 1 + 2g + g^2 \le 1 + 3g \le \kappa.$$

The dyadic bracket supplies the initial rational interval

$$\ell_0 = \frac{r}{2}, \qquad u_0 = r, \qquad \ell_0^2 \le x < u_0^2, \qquad u_0 - \ell_0 = \ell_0.$$

Compute the natural number

$$N = \left\lceil \frac{2}{g} \right\rceil.$$

The finite geometric-depth estimate of lemma 2.7 gives $(2/3)^N \le g$, and hence $(1/2)^N \le g$. Perform exactly N rational bisections. At each step compare x with the square of the midpoint and retain the half with endpoints ℓ_j, u_j satisfying $\ell_j^2 \le x < u_j^2$. Thus

$$u_N - \ell_N = 2^{-N}(u_0 - \ell_0) = 2^{-N}\ell_0 \le g\ell_N,$$

where the last inequality also uses $\ell_0 \le \ell_N$. Consequently

$$u_N \le (1+g)\ell_N, \qquad x < u_N^2 \le (1+g)^2\ell_N^2 \le \kappa x.$$

Taking $\rho = u_N$ proves the requested bracket. Every recursion depth and every comparison has been computed from the displayed rationals before the finite recursion begins.

The threshold makes $\kappa(R/\mu) < s^2$, so $\rho < s$. Moreover $R < \mu\rho^2$, while $\mu \le \eta(B)$ and $\mu \le 2L_1(f, B)$; hence both residual seed budgets in theorem 9.5 hold. The same finite Newton–Rouché construction therefore applies. Composing its bounds $\chi^2(q, z) \le 18\rho^2$ and $\mathrm{diag}^2(D) \le 32\rho^2$ with the requested bracket gives the two displayed estimates. □

Formal verification:

`Purelib.Polynomial.Geometry.DyadicBracket.selectWithin;Purelib.Polynomial.Geometry.ResidualChordal.AdjustableLinearResidualChordalResult.chordalLinear`.

Remark 9.7 (why the converse is local and normalized)**.** *A small value of* $\mathrm{N}(f(q))$ *alone does not identify a track or even a relevant ambient box: multiplying* f *by a nonzero Gaussian rational rescales the residual without changing its winding tracks. The converse in theorem 9.5 therefore includes the displayed squarefree Bézout margins and a known containing winding state.*

The protected parent of Definition 9.4 need not carry winding one: the Newton construction uses its quantitative boundary clearance and ambient refinement together with the already supplied squarefree and axis-elimination certificates. This smaller interface permits a global finite classifier.

The classifier removes the need for the caller to choose that parent. If the sample lies outside the coefficient isolation square, leading-term dominance computes a positive lower bound on its residual, so the small-residual branch is impossible. If it lies inside, the outer square itself supplies the protected parent; the resulting local track is then assigned to one of the already constructed global tracks. The three output branches below are just the decidable outcomes of those two tests plus exact evaluation.

Theorem 9.8 (global displayed-residual classifier)**.** *Let f have degree $d \geq 2$ and carry the finite squarefree certificate used by the box theorem. Let $K_f = [-R, R]^2$ be its coefficient-computed isolation square, let $M_f > 0$ be the leading–tail dominance margin of Definition 6.1, and put*

$$U = \mathrm{N}(\mathrm{lc}(f))\,(2R^2)^d, \qquad \Omega(f) = \min\left\{M_f, \frac{M_f^2}{4U}\right\} > 0.$$

For every displayed $q \in \mathbb{Q}(i)$ and every $\varepsilon \in \mathbb{Q}_{>0}$ one can compute a positive rational threshold $\Delta_{\mathrm{glob}}(f, q, \varepsilon)$ and return exactly one of the following finite records:

1. *the exact evaluation certificate $f(q) = 0$;*
2. *a winding-one box $D \subset K_f$, its total shrinking RAT, the estimates*

$$\mathrm{diag}^2(D) \leq 32\varepsilon^2, \qquad \chi^2(q, z) \leq 18\varepsilon^2 \quad (z \in D),$$

   and a deterministic owner of this RAT among the d tracks already returned by global isolation;
3. *the displayed inequality*

$$\Delta_{\mathrm{glob}}(f, q, \varepsilon) \leq \mathrm{N}(f(q))\,,$$

   certifying failure of the explicit small-residual hypothesis.

Inside K_f, the threshold is the uniform protected-parent target from theorem 9.5; outside K_f, it is $\Omega(f)$. In particular, no sample outside K_f can enter the assigned-box branch with residual below the stated threshold.

Proof. Whenever a displayed q satisfies $R^2 \leq \mathrm{N}(q) \leq 2R^2$, Lemma 6.2 gives

$$M_f \leq \mathrm{N}(L_f(q)) - \mathrm{N}(T_f(q))\,.$$

For displayed Gaussian rationals a, b, a two-coordinate Cauchy identity gives the finite inequality

$$0 < m \leq \mathrm{N}(a) - \mathrm{N}(b) \quad \Longrightarrow \quad \frac{m^2}{4\,\mathrm{N}(a)} \leq \mathrm{N}(a + b)\,.$$

On the annulus, $\mathrm{N}(L_f(q)) \leq U$, so $M_f^2/(4U) \leq \mathrm{N}(f(q))$. Beyond the annulus the leading estimate is at least twice the tail estimate. The elementary bound

$$\frac{\mathrm{N}(a)}{2} - \mathrm{N}(b) \leq \mathrm{N}(a + b)$$

then gives $M_f \leq \mathrm{N}(f(q))$. Failure of the closed-square membership predicate implies $R^2 < \mathrm{N}(q)$ by one of four decidable coordinate comparisons, so the two cases prove $\Omega(f) \leq \mathrm{N}(f(q))$ for every displayed q rejected by the membership predicate of K_f.

For $q \in K_f$, use the boundary clearance of K_f as a `ProtectedAmbient`. A residual below the uniform target first proves strict interior membership, then the same Newton–Rouché calculation builds D and its track. The track refines K_f. The finite maximality assignment therefore refines it to the coefficient-computed partition scale and returns the selected output owner; a zero-winding remainder cell is excluded by additivity and nonnegativity. Every branch is selected by exact rational comparisons. □

Formal verification:

`Purelib.Polynomial.Geometry.GlobalResidual.classify;Purelib.Polynomial.Geometry.GlobalResidual.AssignedResult.owner`.

10 Irreducibility and general polynomials

This section constructs multiplicity presentations, applies the squarefree RAT construction to their factors, and transfers the resulting finite observations to the original polynomial.

Definition 10.1 (finite irreducibility certificate)**.** *A* finite irreducibility certificate *for a displayed polynomial $f \in \mathbb{Q}(i)[Z]$ records that f is nonzero and has positive degree, together with the following proper-divisor rule. For every displayed nonzero $h, k \in \mathbb{Q}(i)[Z]$ satisfying $f = hk$ and $\deg h < \deg f$, it certifies $\deg h = 0$.*

Formal verification:

`Purelib.Polynomial.Irreducible.IrreducibleCertificate`.

Proposition 10.2 (irreducibility supplies squarefreeness)**.** *From a finite irreducibility certificate for f, the Euclidean algorithm constructs the squarefree certificate of definition 1.1. If $\deg f > 1$, the same certificate excludes every displayed Gaussian-rational linear factor.*

Proof. Compute $g = \gcd(f, f')$ and the exact factorization $f = gh$. The derivative calculation gives $\deg g \le \deg f' < \deg f$, so the proper-divisor rule makes g constant. Rescaling the computed Bézout cofactors gives $Uf + Vf' = 1$. If $\deg f > 1$, a displayed linear factor would itself be a nonconstant proper divisor, contrary to the same rule. □

Formal verification:

`Purelib.Polynomial.Irreducible.squarefreeOfIrreducible`.

For completeness, arbitrary polynomials can be reduced constructively.

The purpose of this preprocessing is narrower than full factorization. A repeated factor should describe one geometric site carrying a larger integer weight, not several indistinguishable tracks. The algorithm therefore splits only until it has pairwise-coprime squarefree factors and records how often each occurs. The box theorem is then run once per squarefree factor, while the multiplicity is transported as finite integer data.

Lemma 10.3 (squarefree certificate or smaller exact split)**.** *Let $p \in \mathbb{Q}(i)[Z]$ be displayed, normalized, nonzero, and of positive degree. A finite Euclidean calculation returns exactly one of the following records:*

(i) a squarefree certificate for p in the sense of Definition 1.1;

(ii) displayed normalized nonzero polynomials g, q of positive degree, each of degree strictly below $\deg p$*, together with the identities*

$$p = gq, \qquad \deg g + \deg q = \deg p.$$

Proof. Compute the extended Euclidean GCD of p and p'. Positive degree makes p' nonzero and gives $\deg p' < \deg p$, while nonzeroness of p makes the computed GCD g nonzero. Its degree is decidably either zero or positive. If $\deg g = 0$, rescale the returned Bézout identity exactly as in the proof of Proposition 10.2; this gives the squarefree certificate of Definition 1.1.

If $\deg g > 0$, exact Euclidean division returns the normalized cofactor q and the identity $p = gq$. Since g divides p', one has $\deg g \leq \deg p' < \deg p$. Exact degree additivity gives $\deg g + \deg q = \deg p$; hence q is also nonzero, normalized, of positive degree, and of degree strictly below $\deg p$. □

Formal verification:

`Purelib.Polynomial.Squarefree.squarefreeOutcome;Purelib.Polynomial.Factorization.factor_natDegree_add;Purelib.Polynomial.Factorization.decomposeNormalized`.

Lemma 10.4 (finite coprime multiplicity fold)**.** *Let P be a displayed normalized positive-degree polynomial, and suppose a nonempty finite list of normalized positive-degree squarefree occurrences has product P and degree sum* $\deg P$*. A terminating finite GCD fold converts that list into a finite squarefree-multiplicity presentation of P in the sense of Definition 1.19. In particular, the returned positive-degree basis is nonempty, its factors are pairwise coprime with displayed Bézout certificates, and its weighted product and degree identities recover those of the occurrence list.*

Proof. Insert the occurrences successively into an initially empty basis. When a new occurrence is coprime to the current head, retain the head and recurse through the tail. Otherwise compute their nonconstant GCD and replace the overlap by its common factor and the two exact cofactors. Squarefreeness supplies the pairwise-coprimality certificates among those three parts and preserves coprimality with the unvisited tail. Every recursive insertion consumes one entry of the displayed finite list.

The insertion invariant records pairwise coprimality, the weighted product identity, and the weighted degree identity. After the fold, remove only degree-zero basis parts and multiply their displayed values into one nonzero unit. The input has positive degree, so the remaining positive-degree basis is nonempty. These invariants are precisely the data required by Definition 1.19. □

Formal verification:

`Purelib.Polynomial.Multiplicity.insertPart;Purelib.Polynomial.Multiplicity.buildBasis;Purelib.Polynomial.Factorization.Decomposition.positiveMultiplicityProductIdentity;Purelib.Polynomial.Factorization.Decomposition.positiveMultiplicityEntries_pairwise;Purelib.Polynomial.Factorization.Decomposition.positiveMultiplicityEntries_nonempty;Purelib.Polynomial.Factorization.Decomposition.positiveMultiplicityDegree`.

Proposition 10.5 (finite squarefree and multiplicity preprocessing)**.** *There is a total coefficient algorithm which distinguishes the zero polynomial, a nonzero constant, and a positive-degree polynomial. In the positive-degree case it returns a nonempty finite list of normalized, positive-degree, squarefree occurrences whose product is the normalized input and whose degrees sum to its degree. It also returns the resulting finite squarefree-multiplicity presentation in the sense of Definition 1.19.*

Proof. The normalized coefficient list decides whether the input is zero, a nonzero constant, or of positive degree. In the positive-degree branch, repeatedly apply Lemma 10.3. Retain a returned squarefree factor; in the split branch recurse on the two strictly smaller natural degrees and append their occurrence lists. Polynomial multiplication and degree additivity preserve the two stated invariants, and recursion terminates by the strict degree decrease. Apply Lemma 10.4 to the resulting nonempty occurrence list to obtain the presentation of Definition 1.19. □

Formal verification:

`Purelib.Polynomial.Factorization.preprocess`.

Run the box theorem once for each positive-degree basis factor and attach its displayed multiplicity. Repeated factors therefore do not create duplicate box machines. This preprocessing is intentionally not claimed to be an irreducible factorization algorithm; no such algorithm is needed for the squarefree box theorem proved here.

The next theorem has two complementary readings. If the value of a finite product is very small, a bounded scan can exhibit at least one small factor. If a particular factor has already been selected and the product of the others has a positive lower bound, then smallness can be transferred back and forth with the selected factor, with its multiplicity appearing as an ordinary natural power. No radical is needed to undo that power.

Theorem 10.6 (finite multiplicity residual transfer)**.** *Write the computed positive multiplicity basis as*

$$p(q) = u \prod_{j=1}^{s} h_j(q)^{m_j}, \qquad u \neq 0, \quad m_j \in \mathbb{N}_{>0},$$

where this is an equality of displayed Gaussian-rational evaluations. Put $U_0 = \mathrm{N}(u) > 0$ *and* $N = \sum_j m_j$*. For every displayed* $\delta \in \mathbb{Q}_{>0}$*,*

$$\mathrm{N}(p(q)) < U_0 \delta^N$$

constructively selects an index j *with* $\mathrm{N}(h_j(q)) < \delta$*.*

For a preselected entry $h = h_j$ *of multiplicity* $m = m_j$*, expose the finite list split before and after that entry and let* $c(q)$ *be the displayed product of all the other positive factors. If a displayed certificate gives*

$$0 < \beta \leq U_0 \,\mathrm{N}(c(q))$$

and a local residual theorem gives $0 \leq X \leq \kappa \,\mathrm{N}(h(q))$ *with* $0 \leq \kappa$*, then*

$$X^m \leq \left(\kappa \max\{1, \beta^{-1}\}\right)^m \mathrm{N}(p(q)) \,. \tag{23}$$

In particular X *may be the affine Hermitian projective quadrance supplied by the requested-factor bound of corollary 9.6, with its displayed coefficient used as* κ *here. The result uses the natural power* m*; it does not extract an* m*-th root or introduce a fractional exponent.*

Proof. Exact norm multiplicativity turns the basis identity into

$$\mathrm{N}(p(q)) = U_0 \prod_j \mathrm{N}(h_j(q))^{m_j} .$$

A bounded scan tests the finitely many inequalities $\mathrm{N}(h_j(q)) < \delta$. If every test failed, monotonicity of natural powers and finite products would give the reverse bound $U_0\delta^N \le \mathrm{N}(p(q))$, a contradiction. The scan therefore returns the first proof-bearing small factor.

For the preselected entry, the exact list split gives

$$\mathrm{N}(p(q)) = U_0\,\mathrm{N}(c(q))\,\mathrm{N}(h(q))^m , \qquad \beta\,\mathrm{N}(h(q))^m \le \mathrm{N}(p(q)) .$$

Raise the local inequality to the positive natural power m. Since $M = \max\{1, \beta^{-1}\} \ge 1$ and $m > 0$, one has $\beta^{-1} \le M \le M^m$. Multiplying the preceding inequalities proves (23). All products and scans are structural recursions on displayed finite lists. □

Formal verification:

`Purelib.Polynomial.MultiplicityResidual.selectSmallFactor;Purelib.Polynomial.MultiplicityResidual.chordalPow_le_integerPowerConstant_mul_sourceResidual`.

11 A finite rational-function argument principle

This section combines a reduced quotient with multiplicity presentations to produce jointly separated signed sites and compute the rational-function action principle.

Let $F, G \in \mathbb{Q}(i)[Z]$ be displayed dense polynomials and suppose that G is nonzero. The identically zero numerator is returned by a separate constructor. In the remainder of this section F is also nonzero.

Conceptually, the construction performs three operations. It first cancels the common polynomial factor, since a displayed quotient has no zero/pole distinction there. It then applies the polynomial box theorem separately to the squarefree factors of the reduced numerator and denominator. Finally it labels numerator sites positively, denominator sites negatively, and uses Bézout certificates to separate every pair across both sides. The outer winding is consequently the positive degree total minus the negative degree total.

11.1 Executable cancellation

Run extended Euclid on the normalized pair. Its finite output supplies displayed polynomials

$$H, \quad F_0, \quad G_0, \quad U, \quad V \in \mathbb{Q}(i)[Z]$$

and the identities

$$F = HF_0, \qquad G = HG_0, \qquad UF_0 + VG_0 = 1. \tag{24}$$

The Bézout identity is the finite certificate that the displayed reduced cofactors have no common nonconstant factor. Here H and G_0 are nonzero, and F_0 is nonzero when F is. The same finite division records prove

$$\deg H + \deg F_0 = \deg F, \qquad \deg H + \deg G_0 = \deg G. \tag{25}$$

At a displayed $z \in \mathbb{Q}(i)$ for which $G(z) \neq 0$, cancellation gives the exact value identity

$$F(z)G(z)^{-1} = F_0(z)G_0(z)^{-1}.$$

No value is assigned at a point where the denominator vanishes.

On any rational boundary clear for both reduced cofactors, the generic winding of definition 1.11, realized explicitly in section 4, applies to F_0/G_0. In particular, theorem 4.11 computes its winding by subtracting the polynomial windings. Cancellation therefore introduces no second, quotient-specific notion of winding.

The coefficient bounds from lemma 2.4 compute a positive radius for each nonzero polynomial beyond which its leading term dominates. Take the maximum of the two radii. The resulting finite symmetric square is clear for both reduced cofactors, so theorems 4.11 and 6.3 give the following result, including constant cofactors.

Corollary 11.1 (outer reduced-degree winding)**.** *There is one coefficient-computed positive rational R such that*

$$\mathrm{Wind}_4(F_0/G_0; K_R) = 4(\deg F_0 - \deg G_0). \tag{26}$$

Proof. Choose R as the maximum of the two coefficient-computed dominance radii. Theorem 6.3 gives the two polynomial windings on K_R; Theorem 4.11 subtracts them, giving (26). □

Formal verification:

`Purelib.RationalFunctions.FunctionArgumentPrinciple.NonzeroReducedInput.outer_rationalWinding_eq_degreeDifference`.

11.2 Signed multiplicity sites

Apply proposition 10.5 separately to the two reduced cofactors. It returns nonzero units and pairwise-coprime squarefree positive-degree factors

$$F_0 = u_+ \prod_{j=1}^{r} p_j^{m_j}, \qquad G_0 = u_- \prod_{k=1}^{s} q_k^{n_k}, \tag{27}$$

with $m_j, n_k > 0$ and

$$\sum_{j=1}^{r} m_j \deg p_j = \deg F_0, \qquad \sum_{k=1}^{s} n_k \deg q_k = \deg G_0. \tag{28}$$

Factors on the same side carry the finite pairwise Bézout certificates from the multiplicity construction. Equation (24) also gives a Bézout certificate between every p_j and every q_k: exact finite factorization transports coprimality to displayed nonzero factors.

Each squarefree factor is handled once. A degree-one factor returns its explicit Gaussian-rational zero with an exact evaluation certificate. It is the only branch in which a zero or pole point is returned. A factor of degree at least two returns exactly its degree many shrinking winding-one tracks from theorem 1.17. Thus the output is a finite family of *sites*; a site is either an exact rational terminal or a box track.

This is the finite rational-function analogue of a geometric zero or pole location. The sign records whether the quotient turns in the numerator or denominator direction, the

multiplicity m records the contribution $\pm 4m$, and the terminal/track alternative says how it is observed: exactly when linear and by repeatable boxes otherwise.

The separation cases now require no additional definition. An exact terminal $q = a + ib$ contributes the degenerate support record (a, a, b, b), whereas a RAT contributes the proper box returned by a finite call. Substituting two degenerate records, one of each kind, or two proper boxes into the four inequalities of Definition 1.16 gives respectively the terminal/terminal, terminal/RAT, and RAT/RAT certificates. Signs and multiplicities do not alter this geometric witness.

Fix a numerator factor p_j and one of its higher-degree tracks. The Bézout evaluation bounds compute one positive rational refinement request small enough to protect the track simultaneously from

$$C_j = u_+ \prod_{\ell \neq j} p_\ell^{m_\ell} \quad \text{and} \quad G_0.$$

On the returned box, C_j and G_0 have positive boundary clearances and zero winding. The finite power and product laws therefore give

$$\mathrm{Wind}_4(p_j^{m_j} C_j / G_0; B) = 4m_j.$$

The presentation gives $F_0 = p_j^{m_j} C_j$. The denominator calculation is symmetric and gives

$$\mathrm{Wind}_4(F_0 / q_k^{n_k} D_k; B) = -4n_k, \qquad D_k = u_- \prod_{\ell \neq k} q_\ell^{n_\ell}.$$

Joint separation is also quantitative and finite. Tracks belonging to one squarefree factor are separated by its polynomial isolation output. For two different factors, refine the first until its residual forces a positive Bézout lower bound for the second, and refine the second until its residual is below that bound. Overlap would contradict the two rational inequalities. For an exact rational terminal and a track, the same lower bound excludes the displayed point from a finite refinement of the track. Two exact terminals of coprime factors are unequal because substituting a common displayed point in their Bézout identity would give $1 = 0$.

Proof of Theorem 1.21. Cancellation, quotient winding, multiplicity preprocessing, local signed winding, and the two degree totals were proved above. Enumerate every focused position in each finite multiplicity list. Store the exact terminal or higher isolation result at that position, its internal separation and atomic ownership data, and all same-side separation witnesses. Then enumerate the finite Cartesian product of numerator and denominator positions and apply the cross-factor Bézout construction just described. Constants contribute empty site families. Finally store (26) in the same result record; definition 1.11 gives the normalized identity in Theorem 1.21. All iterations are over displayed finite lists or natural degrees. □

Formal verification:

`Purelib.RationalFunctions.Sites.prove;Purelib.Papers.RFAP.Paper.rationalFunctionArgumentPrinciple`.

The ownership sentence in theorem 1.21 is deliberately atomic. It would be false to say that every arbitrary nonzero-winding box has exactly one site: a larger rational box may surround several zeros and poles. The proved maximality statement concerns winding-one tracks for one squarefree factor. The returned finite partition has zero-winding remainder cells for that factor, and every competing atomic track is assigned to one output cell.

Corollary 11.2 (squarefree signed specialization)**.** *Suppose both nonzero reduced cofactors F_0 and G_0 are supplied with finite squarefree certificates. The RFAP algorithm has exactly* $\deg F_0$ *numerator sites and* $\deg G_0$ *denominator sites. Constants contribute no site, degree-one cofactors contribute exact Gaussian-rational terminals, and higher cofactors contribute degree-indexed box tracks. Every numerator track has winding $+4$, every denominator track has winding -4, the two families are jointly separated, and* (26) *holds.*

Proof. Apply the squarefree box theorem directly to each reduced cofactor instead of multiplicity preprocessing. Protect each higher track from the opposite cofactor using their reduced Bézout certificate. The opposite winding is zero, so theorem 4.11 gives $4-0$ on a numerator track and $0-4$ on a denominator track. The degree-one and constant branches, joint separation, atomic ownership, and outer winding use the constructions already described. □

Formal verification:

`Purelib.RationalFunctions.FunctionArgumentPrinciple.proveSquarefree;`
`Purelib.Papers.RFAP.Paper.squarefreeRationalFunctionArgumentPrinciple`.

11.3 Two-way metric and residual calibration

The word here is *residual*, not the meromorphic residue. For displayed P, Q and a displayed q with $Q(q) \neq 0$, put

$$\mathcal{R}_{P/Q}(q) = \mathrm{N}(P(q)/Q(q)) = \frac{\mathrm{N}(P(q))}{\mathrm{N}(Q(q))}. \tag{29}$$

There are again two directions. Moving far enough down a numerator track makes this quotient residual uniformly small because the numerator becomes small while the protected denominator stays bounded away from zero. Conversely, a sufficiently small quotient residual makes the numerator residual small, because the denominator also has an explicit upper bound; the polynomial converse can then build a nearby numerator track. At a denominator site the same statement is applied to the reciprocal, so its smallness is the finite certificate for the original quotient's displayed blow-up.

Definition 11.3 (local quotient context and protected state)**.** *For displayed coprime polynomials P, Q with* $\deg P > 0$ *and a fixed ambient box A, a* local quotient context *consists of a RAT for P, a current winding-one state on that RAT, and a displayed Bézout certificate $UP + VQ = 1$. Finite refinement using this identity produces its* protected state*. The same coefficient calculations provide positive rationals*

$$0 < \eta_Q \leq \mathrm{N}(Q(z)) \leq M_Q$$

for every displayed z in that state or any refinement of it. The lower bound is the Bézout protection margin and the upper bound is a Horner majorant on A. At a denominator site, interchange P and Q; the source track is then a track for Q protected from P.

Formal verification:

`Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext;Purelib.`
`RationalFunctions.ResidualChordal.LocalQuotientContext.protectedState;`
`Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext.`
`oppositeMargin;Purelib.RationalFunctions.ResidualChordal.`
`LocalQuotientContext.oppositeMajorant`.

Theorem 11.4 (two-way RFAP residual bounds)**.** *Let B be a rational box carrying any winding-one state for P that refines the protected state inside the fixed ambient box A.*

(i) Every displayed $q \in B$ satisfies the forward estimate

$$\mathcal{R}_{P/Q}(q) \le \frac{4L_1(P,A)}{\eta_Q}\operatorname{diag}^2(B). \tag{30}$$

Thus a numerator track makes the reduced rational function small. On a denominator track, with P, Q interchanged, the same estimate makes the reciprocal rational function small.

(ii) Assume P is squarefree of degree at least two, q satisfies the strict interior predicate of B, and $\varepsilon > 0$. Let $\Delta(P, B, q, \varepsilon)$ be the polynomial target from equation (22). The positive quotient target

$$\Delta_{P/Q}(B, q, \varepsilon) = \frac{\Delta(P, B, q, \varepsilon)}{M_Q} \tag{31}$$

has the following constructive converse: if $\mathcal{R}_{P/Q}(q) < \Delta_{P/Q}(B, q, \varepsilon)$, exact evaluation either returns $P(q) = 0$, or constructs a winding-one box $D \subset B$, its total RAT, and

$$\operatorname{diag}^2(D) \le 32\varepsilon^2, \qquad \chi^2(q, z) \le 18\varepsilon^2 \qquad (z \in D).$$

(iii) For each $\varepsilon > 0$, a bounded rational refinement of the protected P-track constructs a state on a box B_ε on which

$$\mathcal{R}_{P/Q}(q) < \varepsilon^2 \qquad (q \in B_\varepsilon). \tag{32}$$

In the pole orientation, $P = G_0$ and $Q = F_0$. Exact evaluation either returns the displayed pole certificate $G_0(q) = 0$, or else

$$\mathrm{N}(F_0(q)/G_0(q)) \ge \frac{1}{\varepsilon^2}. \tag{33}$$

All constants and refinement requests are computed from displayed coefficients, boxes, and Bézout data.

Proof. The polynomial estimate in theorem 1.17 gives $\mathrm{N}(P(q)) \le 4L_1(P, A)\operatorname{diag}^2(B)$, while protection gives $\eta_Q \le \mathrm{N}(Q(q))$; division proves (30). Conversely, (29), the upper bound $\mathrm{N}(Q(q)) \le M_Q$, and (31) give

$$\mathrm{N}(P(q)) = \mathcal{R}_{P/Q}(q)\,\mathrm{N}(Q(q)) < \frac{\Delta(P, B, q, \varepsilon)}{M_Q} M_Q = \Delta(P, B, q, \varepsilon).$$

Apply theorem 9.5.

For (iii), refine the P-track at the explicit uniform residual request with polynomial target $\eta_Q\varepsilon^2$. Every displayed point in the returned box has $\mathrm{N}(P(q)) < \eta_Q\varepsilon^2$ and $\mathrm{N}(Q(q)) \ge \eta_Q$, proving (32). If both displayed values are nonzero, exact norm multiplicativity gives

$$\mathcal{R}_{P/Q}(q)\mathcal{R}_{Q/P}(q) = 1.$$

Inverting the strict upper bound yields the weak lower bound (33). If $P(q) = 0$, the algorithm reports that exact finite evaluation instead of assigning a value at the pole. □

Formal verification:

`Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext.quotientResidualSq_le_diagonal;Purelib.RationalFunctions.ResidualChordal.LocalQuotientContext.oppositeQuotientSq_ge_inv_epsilonSq`.

Example 11.5 (coprime squarefree quotient)**.** *For $F(Z) = Z$ and $G(Z) = Z - 1$, the identity*

$$1 = F(Z) - G(Z)$$

is a finite coprimality certificate. Both derivatives are the constant one. The two sites are therefore the exact displayed points 0 and 1, carrying the exact numerator-zero and denominator-zero evaluations respectively; the outer winding is $4 - 4 = 0$. Their cross-separation witness is the rational coordinate gap 1.

Example 11.6 (repeated factor)**.** *Take $F(Z) = (Z - 1)^2$, $G(Z) = 1$, and the box $B = (-2, 2, -2, 2)$. The finite product calculation gives*

$$\mathrm{Wind}_4(F/G; B) = \mathrm{Wind}_4(Z - 1; B) + \mathrm{Wind}_4(Z - 1; B) - \mathrm{Wind}_4(1; B) = 8.$$

This is proved from two explicit boundary-clearance certificates, the finite product formula, and theorem 4.11. It demonstrates why cancellation alone does not imply the ± 4 specialization and why a repeated geometric site carries multiplicity two rather than being duplicated.

12 Finite size of the RFAP-I calculation

This section bounds the finite arithmetic used to produce one requested refinement state. In the glossary's terms, it covers coefficient clearing, the two eliminants and their Bézout data, the safe cuts, and every Sturm winding used by repeated subdivision. The purpose is not an optimized complexity classification. It is the more basic constructive assertion that the sizes of the intermediate displayed records, as well as the number of calculation clauses, are bounded before the calculation is run.

For this purpose the executable route carries an ordinary rational as a coprime numerator and positive denominator. Before addition it removes the gcd of the two denominators; before multiplication it cross-cancels; after each operation it performs a final constructive gcd reduction. Gaussian rationals are pairs of these reduced records. These canonical records, and the normalized polynomials, matrices, and chains built from them, are the active algebraic objects used in the proofs above. Structural equality of canonical fractions is equivalent to the displayed cross-multiplication identity, while normalization returns an explicit certificate of that identity. No erasure into a quotient-valued rational is part of the paper-facing proof route.

This reduced-fraction layer is slightly more infrastructure than the headline RFAP-I theorem alone requires. We have embedded it in the Lean development partly to enable the companion approximation work, where bounds on the numerators and denominators carried by actual calculations become mathematical output rather than merely an implementation concern. Keeping the reduced records and their normalization certificates lets that later work reuse the RFAP-I algorithms through the same stable canonical algebraic interface, without assuming that a small rational value has a small unreduced representative.

Write $\ell(n) = 1 + \lfloor \log_2 n \rfloor$ for $n > 0$, with $\ell(0) = 1$. For a reduced rational record, its component magnitude is the maximum of the absolute numerator and denominator; for a Gaussian record it is the maximum over both coordinates. Polynomial and matrix magnitudes are the corresponding maxima over their finite coefficient lists. All uses of ℓ below are implemented by recursion on natural numbers, not by real logarithms.

Definition 12.1 (RFAP-I calculation budgets)**.** *Fix a displayed polynomial f, a proper boundary-clear winding state on a rational box, its displayed vertical and horizontal elimination certificate, and a positive rational refinement request. Let r be the geometric depth computed from that request.*

The magnitude recurrence M is the maximum of the following source recurrences:

1. *the coefficient recurrence for the vertical separated-Sylvester matrix and its complete fraction-free Schur inverse;*
2. *the same recurrence after the quarter turn used for the horizontal eliminant;*
3. *the product denominator and cleared-coefficient envelope of the input coefficient list;*
4. *the normalized safe-cut recurrence applied r times to the four endpoints of the initial box;*
5. *the normalized coefficient and candidate-evaluation envelope for the two displayed elimination polynomials; and*
6. *the coefficient and endpoint-evaluation envelopes of all normalized Sturm chains used for the four child windings in those r rounds.*

Put $B = \ell(M)$. The step budget T is the sum of the displayed coefficient-clearing count, the two finite Schur-recursion counts, the two bounded candidate scans and child-selection count in every round, and the field-operation budgets of those child windings. Both T and B are primitive finite recurrences in the supplied records.

Theorem 12.2 (RFAP-I calculation-size certificate)**.** *For every input to definition 12.1, the computed numbers T and B have the following properties.*

1. *Coefficient clearing, both separated-Sylvester–Schur calculations, all safe-coordinate scans, and all subdivision windings fit their respective summands of T.*
2. *Every coefficient of both normalized separated-Sylvester scalars and every polynomial entry of their scaled inverses has component width at most B. In particular this covers the finite polynomial data from which the displayed Bézout cofactors are extracted.*
3. *The common coefficient scale and both integral components of every cleared coefficient have width at most B.*
4. *Every endpoint of the retained box after the requested refinement has width at most B.*
5. *For every child box examined during refinement, every coefficient of every normalized signed-remainder-chain member and every normalized endpoint evaluation used by its four edge indices has width at most B.*

The normalized chains compute exactly the headline Cauchy indices, so the bounded execution produces the same windings and retained refinement state as the construction proved above.

Proof. One normalized field operation has an explicit component recurrence. Horner induction therefore bounds evaluation of a polynomial with bounded coefficients and length. Long-division induction applies the same recurrence to every signed-remainder-chain member; its degree drops strictly at each recursive call. Fraction-free Schur induction bounds the current matrix and then every entry of the recursively assembled scaled inverse. These

two inductions also retain coefficient-list bounds, so their evaluation envelopes are source-computed rather than maxima of completed traces.

Each safe-coordinate search tests exactly $d_0 + 1$ candidates, where d_0 is the supplied eliminant degree bound. The selected coordinate is one of those candidates, and its reduced numerator and denominator satisfy the same fixed operation envelope. Induction on r carries this bound through every retained child. Summing the finite clause bounds gives T; inclusion in the nested maximum M and monotonicity of ℓ give B. Finally, the erasure theorems commute with polynomial arithmetic, long division, evaluation, and the Schur construction. They identify the normalized execution with the coefficientwise identities and windings already used by the paper-facing theorems. □

Formal verification:

`Purelib.RootIsolation.Sturm.Normalized.erase_canonical_signedRemainderChain;Purelib.RootIsolation.Sturm.Normalized.chainMember_resource;Purelib.Polynomial.NormalizedEliminant.erase_normalizedEliminant;Purelib.Polynomial.NormalizedBezoutResource.normalizedSeparatedData_within;Purelib.Polynomial.ClearingCalculationSize.scaledGaussianCoefficient_component_le;Purelib.RootIsolation.Isolation.CalculationSize.refineTo_box_within;Purelib.RootIsolation.CalculationSize.stageThreeStepBudget;Purelib.RootIsolation.CalculationSize.stageThreeBitBudget;Purelib.Papers.RFAP.Paper.stageThreeResourceCertificate.`

The estimates deliberately prefer transparent uniform recurrences to tight ones. Fraction-free subresultant estimates, a common subdivision grid, or a more detailed machine-cost model could improve them. None is needed for the present conclusion: every named intermediate record on the requested RFAP-I path has a displayed finite width bound.

13 Exact cubic example: $Z^3 - (1 + i)$

This section traces one displayed polynomial through its squarefree certificate, coefficient isolation square, box winding, and winding-one boxes.

Consider

$$f(Z) = Z^3 - (1 + i).$$

Its normalized high-to-low coefficient list and derivative list are

$$[a_3, a_2, a_1, a_0] = [1, 0, 0, -(1 + i)], \qquad [f'] = [3, 0, 0].$$

It is Eisenstein in $\mathbb{Z}[i][Z]$ at the Gaussian prime $1 + i$: the two intermediate coefficients and the constant term are divisible by $1 + i$, the constant term is not divisible by $(1 + i)^2$, and the leading coefficient is not divisible by $1 + i$. Thus it is irreducible in the ordinary algebraic terminology.

The box construction itself needs only an exact squarefree certificate. Take the two polynomials

$$U(Z) = \frac{-1 + i}{2} = -\frac{1}{1 + i}, \qquad V(Z) = \frac{1 - i}{6} Z = \frac{Z}{3(1 + i)}.$$

The coefficient of Z^3 in $Uf + Vf'$ and its constant coefficient are respectively

$$\frac{-1+i}{2} + 3\frac{1-i}{6} = 0, \qquad -\frac{-1+i}{2}(1+i) = 1,$$

while every other coefficient is zero. Hence the displayed coefficientwise identity is $Uf + Vf' = 1$.

We next run the coefficient construction of definition 1.6. Since

$$C_f = 3\big(\mathrm{N}(-(1+i)) + 2\,\mathrm{N}(0) + 4\,\mathrm{N}(0)\big) = 6, \qquad R_f = \max\{1, 6/\mathrm{N}(1) + 1\} = 7,$$

the coefficient isolation square is $K_f = K_7 = (-7, 7, -7, 7)$. Its exact leading–tail margin is

$$M_f = (\mathrm{N}(1) \cdot 7^2 - C_f)(7^2)^{3-1} = (49 - 6)49^2 = 103243 > 0.$$

The single global application of theorem 6.3 therefore gives $\mathrm{Wind}_4(f; K_f) = 12$.

The coefficient polynomials actually passed to the eliminant and edge calculations are obtained by expanding

$$f(x+iy) = A_f(x,y) + iB_f(x,y), \qquad A_f = x^3 - 3xy^2 - 1, \quad B_f = 3x^2y - y^3 - 1.$$

For example, on a horizontal edge put $x(t) = \alpha + (\beta - \alpha)t$ and $y(t) = c$; the two input lists are the coefficients of

$$x(t)^3 - 3c^2x(t) - 1, \qquad 3cx(t)^2 - c^3 - 1.$$

On a vertical edge one instead puts $x(t) = c$ and $y(t) = \alpha + (\beta - \alpha)t$, obtaining

$$c^3 - 3cy(t)^2 - 1, \qquad 3c^2y(t) - y(t)^3 - 1.$$

The separated Schur–Sylvester recursion on these coefficient data constructs the two axis eliminants and computes the common finite degree bound $D_f = 168$.

The local coefficient bounds are equally explicit. On K_7 the rational radius parameter in definition 7.1 is $R_{K_7} = 98$, and the Horner triples are

$$(0,0,0),\ (2,0,0),\ (392,4,0),\ (76832,1568,8),\ (15059076,460992,4704).$$

Consequently $L_1(f, K_7) = 460992$ and $L_2(f, K_7) = 4704$. Applying the same finite evaluation scan to the displayed cofactors gives

$$M_U = 1, \qquad M_V = \frac{196}{9}, \qquad \tau(K_7) = \frac{1}{16}, \qquad \eta(K_7) = \frac{9}{3136}.$$

Thus the principal numerical outputs of the global and local coefficient stages are explicit; the degree-168 eliminant lists themselves are produced by the stated finite recursion rather than typeset.

A certificate-level execution of the safe subdivision records the following three dyadic winding-one boxes. The edge columns display the four Cauchy–Sturm contributions in positive boundary order; their sum is four in every row. Each contribution uses the factor-two convention fixed in definition 3.9. This is where the direct edge calculations are informative; repeating them on K_f would merely recompute the global winding already supplied by theorem 6.3.

track	rational box $[x_-, x_+] \times [y_-, y_+]$	edge indices	$\mathrm{Wind}_4(f; B)$
1	$[138/128, 140/128] \times [36/128, 38/128]$	$(0, 2, 0, 2)$	4
2	$[-102/128, -101/128] \times [101/128, 102/128]$	$(2, 0, 2, 0)$	4
3	$[-38/128, -36/128] \times [-140/128, -138/128]$	$(0, 2, 0, 2)$	4

For one concrete view of the coefficient arithmetic behind the table, the bottom edge of the first box has $x(t) = (69 + t)/64$ and $y(t) = 9/32$. After clearing the common positive denominator 64^3, its two Cauchy–Sturm inputs are

$$A_e(t) = t^3 + 207t^2 + 13311t - 703, \qquad B_e(t) = 54t^2 + 7452t - 10882.$$

The finite signed-remainder calculation gives contribution zero, the first entry in that row. The other eleven edge pairs are obtained by the same two substitution formulas above.

Taking the rational centers gives

$$q_1 = \frac{139 + 37i}{128}, \qquad q_2 = \frac{-203 + 203i}{256}, \qquad q_3 = \frac{-37 - 139i}{128}.$$

Exact evaluation at these rational samples gives

$$\mathrm{N}(f(q_1)) = \mathrm{N}(f(q_3)) = \frac{39952889}{549755813888} \approx 7.27 \times 10^{-5},$$
$$\mathrm{N}(f(q_2)) = \frac{537358761}{35184372088832} \approx 1.53 \times 10^{-5}.$$

The first polynomial track admits the following entirely finite refinement trace.

state	winding-one box	$\mathrm{diag}^2(B)$
0	$[69/64, 35/32] \times [9/32, 19/64]$	1/2048
1	$[555/512, 1111/1024] \times [297/1024, 149/512]$	1/524288
2	$[71055/65536, 4441/4096] \times [19039/65536, 595/2048]$	1/2147483648

The edge-index certificates are respectively $(0, 2, 0, 2)$, $(0, 2, 0, 2)$, and $(2, 0, 0, 2)$, so every row has winding four. Their rational centers are nested approximants with residual norms

$$\frac{39952889}{549755813888} \approx 7.27 \times 10^{-5},$$
$$\frac{8767783105145}{9223372036854775808} \approx 9.51 \times 10^{-7},$$
$$\frac{546115072853076822625}{6338253001141147007483516026 88} \approx 8.62 \times 10^{-10}.$$

These are successive finite refinement states.

14 Conclusion

A squarefree linear polynomial has a directly computed exact Gaussian-rational zero. In degree $d \geq 2$, the polynomial theorem returns exactly d separated interfaces that answer every finite refinement request. Each finite observation consists of rational boxes, rational witnesses, and Sturm/eliminant certificates. The same squared-diagonal budget bounds the anchor's Hermitian projective quadrance and, up to the explicit factor $L_1(f, A)$, its

polynomial residual quadrance. The global count is four times the degree, local windings are four, and distinctness has a coefficient-computable separation radius. Exact terminal outputs carry a Gaussian-rational evaluation certificate.

The RFAP addendum preserves the same discipline. It cancels a displayed quotient by finite Euclid, attaches signed multiplicities to exact rational terminals or independently shrinking tracks, separates numerator and denominator sites using finite Bézout margins, and computes the outer winding as four times the reduced degree difference. Repeated factors remain one geometric site with a larger winding; the squarefree specialization has only $+4$ and -4 sites.

Every requested approximation is produced in finite time by the total finite procedures proved above. Moreover, theorem 12.2 computes in advance a finite operation budget and component-width budget for the eliminant, clearing, Sturm, winding, safe-cut, and requested-refinement calculations that produce it.

A Formalization and dependency audit

Our use of Lean should not be read as unqualified approval of its language design. Its small proof-checking kernel is valuable; its authoring language is often remarkably unpleasant. Much of the surface feels accumulated rather than designed: dependent syntax, macros, tactic dialects, coercions, typeclass search, and elaboration heuristics overlap in ways that are ad hoc and difficult to predict. A mathematically transparent step can produce source text whose meaning depends on an invisible elaboration path, and a failed step can produce diagnostics describing the elaborator rather than the mathematical obstruction. The result is too often noisy, indecipherable, and unnecessarily ugly. Although users can introduce local notation, they cannot turn the system as a whole into a language-neutral mathematical vernacular: core commands, library names, and diagnostics remain overwhelmingly English. It is difficult to reconcile these choices with the mathematical tradition that good notation should expose structure with economy and beauty. This is not merely an aesthetic complaint. Obscure elaboration makes proofs harder to audit and makes unintended dependencies easier to miss.

Mathlib presented the more consequential difficulty. Its breadth is an extraordinary resource, but many convenient general-purpose routes carry `Classical.choice` into proofs of statements whose data and content are entirely finite. An import by itself does not put choice into the axiom closure of a theorem. The problem arises when an apparently elementary lemma, bundled structure, normalization routine, or synthesized instance uses a choice-bearing route, which then appears in `#print axioms` for the paper-facing result. For ordinary classical developments this may be harmless. Here it would silently surrender exactly the claim under investigation. A machine-checked theorem is not a choice-free theorem merely because the kernel accepts its proof term; its complete dependency closure must also be choice-free.

We therefore audited actual proof dependencies and replaced the offending routes rather than accepting them as library overhead. The project-owned `Purelib.*` modules provide explicit field laws, constructive fractions and Gaussian rationals, executable dense polynomial arithmetic and Euclid, finite bivariate and matrix calculations, and direct separated-Sylvester eliminant certificates. Elsewhere we replaced generic order, inverse, normalization, and finite-cardinality arguments by direct sign proofs and structural recursion on displayed finite data. Some of these local arguments prove facts that are elementary on

paper; their length in Lean records how difficult the library made it to use those facts without an irrelevant selection principle.

The remaining dependence on propositional extensionality was no more indispensable. Simplifier-generated proposition equalities, constructor injectivity certificates, and inherited Core proofs could introduce `propext` into otherwise direct finite arguments. We replaced those routes by explicit implications, equality elimination, structural induction, and proof-producing certificates, including tracked repairs to the Core providers. Finite arithmetic does not become extensional merely because a simplifier chose to rewrite a proposition instead of using an implication. Nor should users have to patch foundational library proofs to avoid axioms that the mathematical argument never needed.

The repository-wide audit now reports an empty transitive axiom closure for every selected mathematical declaration: neither `propext`, `Classical.choice`, `Quot.sound`, nor any other axiom appears. The selection includes private mathematical declarations, registered compiler rewrites, and active script entrypoints. Every selected declaration's full dependency closure is checked, including its generated proof helpers. Independent compiler and elaborator helpers outside those closures are a separate audit obligation; clean mathematical certificates do not establish that their generators are axiom-free. An independent nanoda check accepts the complete exported proof dependencies under the same empty-axiom policy.

The earlier fraction implementation still exposed the axiom `Quot.sound`. It did so only after an explicit cross-multiplication certificate had proved that two displayed representatives were equivalent. That is not a selection principle, but it still obscured the finite justification behind ambient quotient equality. We therefore replaced quotient fractions by computably canonical reduced records. Equivalence and normalization certificates remain first-class evidence, and equality of canonical forms is proved directly. The same discipline required structural representations of the other finite objects and direct equations for several recursive algorithms. The former implementation is retired and remains available in Git history; no active paper declaration imports it. The result answers the interface question constructively: whenever canonicalization is available, the kernel need not be asked to identify quotient classes at all.

There is a broader library-design point. Classical mathematics belongs in a general mathematical library, but unnecessary axioms should not be baked into results that admit finite direct proofs. When the same conclusion can be established without those axioms, removing them is a genuine strengthening of the theorem, not a stylistic variant. Constructive primitives should sit below optional classical conveniences, and axiom dependencies should be visible enough that users can preserve such a boundary without rebuilding basic algebra. Replacing a gratuitous assumption by a proof that does not need it is an improvement that a foundational library should actively prefer. We offer the local replacements in this development as evidence that this stronger organization is practical.

Reproducibility. The `papers.RFAP.RFAP` module exposes the principal executable declarations in the `Purelib.Papers.RFAP.Paper` namespace:

```
Purelib.Papers.RFAP.Paper.isolateByBoxes
Purelib.Papers.RFAP.Paper.isolateGeneralByBoxes
Purelib.Papers.RFAP.Paper.rationalFunctionArgumentPrinciple
Purelib.Papers.RFAP.Paper.squarefreeRationalFunctionArgumentPrinciple
Purelib.Papers.RFAP.Paper.stageThreeResourceCertificate
```

From the root of the accompanying source archive, the complete reproducibility check is

```
scripts/run-lake.sh --wfail build
scripts/run-lake.sh --wfail build PurelibExamples Test \
  CCLExamples DivineProportionsExamples \
  RationalRotationsExamples
scripts/check-axioms.sh --configured-scripts --full
scripts/check-nanoda.sh
```

The wrapper selects the pinned toolchain with its tracked Core proof repairs. The warning-fatal builds check the reusable and publication sources together with the retained tests and optional examples. The full axiom checker also builds and audits configured script owners; it requires empty complete dependency closures for its selected declarations. The final command independently checks the complete exported proof dependencies from every active source module while admitting no axiom declarations. The repository README documents toolchain preparation and the exact audit-root policy.

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